The Mechanism of the Motion of Matter from the Bost–Connes System

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Abstract

We investigate how the same matter changes position while undergoing continual maintenance, using the dynamics of a finite Bost–Connes state space. Formed matter and the accompanying degrees of freedom involved in its maintenance constitute a complete carrier. Internal action and position exchange jointly preserve this carrier while changing its position amplitudes and relative phases, with local position currents composing the propagation of the wave packet as a whole. In the uniform propagation model used here, with time calibrated by the total elapsed duration, the internal mass action determines both the rest energy and the curvature of the propagation spectrum. The long-wavelength, low-momentum expansion of a massive branch gives $M_{\mathrm{in}}c_0^2 = E_0$, linking the continual maintenance of matter to its inertial response. The same propagation structure constrains the position and momentum variances of localized preparations and recovers the uncertainty principle in a controlled continuum limit.

The observable manifestations of motion further depend on the records formed during propagation. The double-slit cross term is controlled by the overlap of the complete unread outputs of the two paths; interference in the unconditioned screen distribution disappears when orthogonal records preserve the path distinction. Surviving matter continues to propagate from the conditional state following a record. When a composite object's collective branch is preserved, its conditional position distribution remains concentrated, and spreading and recoil are controlled, the low-speed equation for its center, together with successive records, gives an approximate description of a macroscopic trajectory. The maintenance of matter, coherent quantum propagation, and macroscopic motion are thus connected within the same dynamics.

Keywords: Bost–Connes system; matter motion; inertial mass; quantum instrument; interference; macroscopic trajectory

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§1 Introduction

Matter motion is the spatial propagation of the same matter under continual maintenance. Its internal structure is preserved, while the amplitudes at different positions and their coherence relations change under the actual action. Exchanges between neighboring positions compose the propagation of the entire matter state. Its joint evolution with other matter, the background, and apparatus makes this propagation manifest as comparable phenomena of motion.

Dynamics of the Bost–Connes State Space [1] establishes motion orientation and displacement, coherent round trips, and position readout. *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System* [2] constructs matter and its accompanying resources through actual formation processes. It shows how the complete carrier is preserved through continuing returns, how its mass phase arises, and how this connects to position propagation of the same carrier. Taking this formed matter as our starting point, we develop global propagation from local position currents and study inertial response, exchange, and records of motion.

Locally, the amplitude at a given position is acted on simultaneously by internal maintenance and exchange with neighboring positions. Internal maintenance acts on the motion orientation at that position, while neighboring components jointly change its subsequent amplitude through coherent exchange. The selected matter mode is preserved, and the relation between orientation and position continues to evolve.

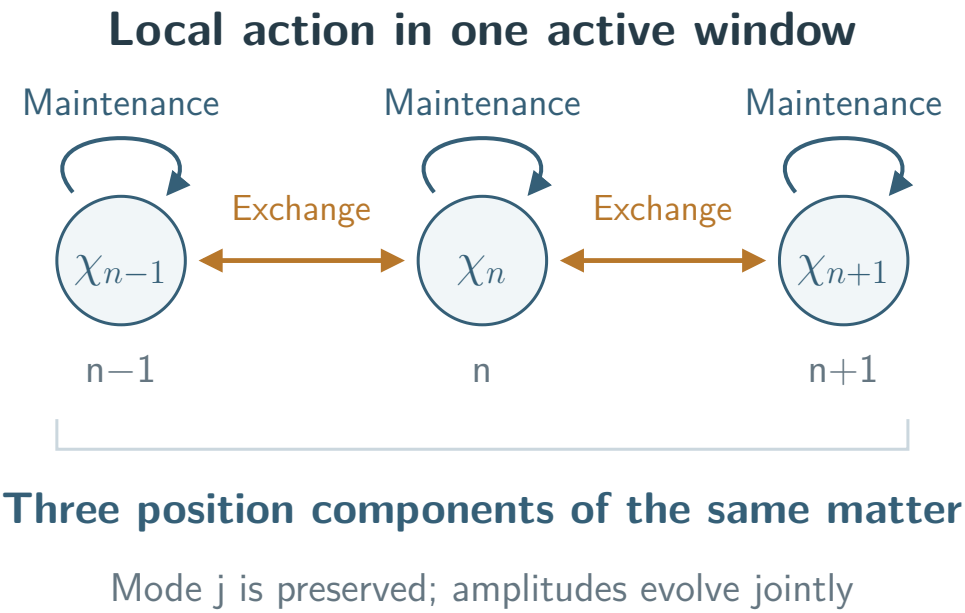


Figure 1-1. **Local action in motion.** χ_{n-1}, χ_n, and χ_{n+1} are the motion-orientation amplitudes of the same matter at three neighboring positions. Curved arrows denote internal action, and horizontal edges denote coherent amplitude exchange; both occur within the same active window. The three circles are position components of one state. The selected matter mode j is preserved while the amplitudes continue to evolve.

Applying this local relation at every position gives the global propagation of the same matter. The change at each position depends on the amplitudes and phases at neighboring positions; all components evolve together to form a time-dependent position distribution and position current. A spatially concentrated wave packet may move, spread, or split, with its specific form determined jointly by the initial preparation and the propagation action. Local flow and the global distribution belong to the same evolution. Even when the center remains at rest, the position current and the shape of the distribution must still be determined from the actual state.

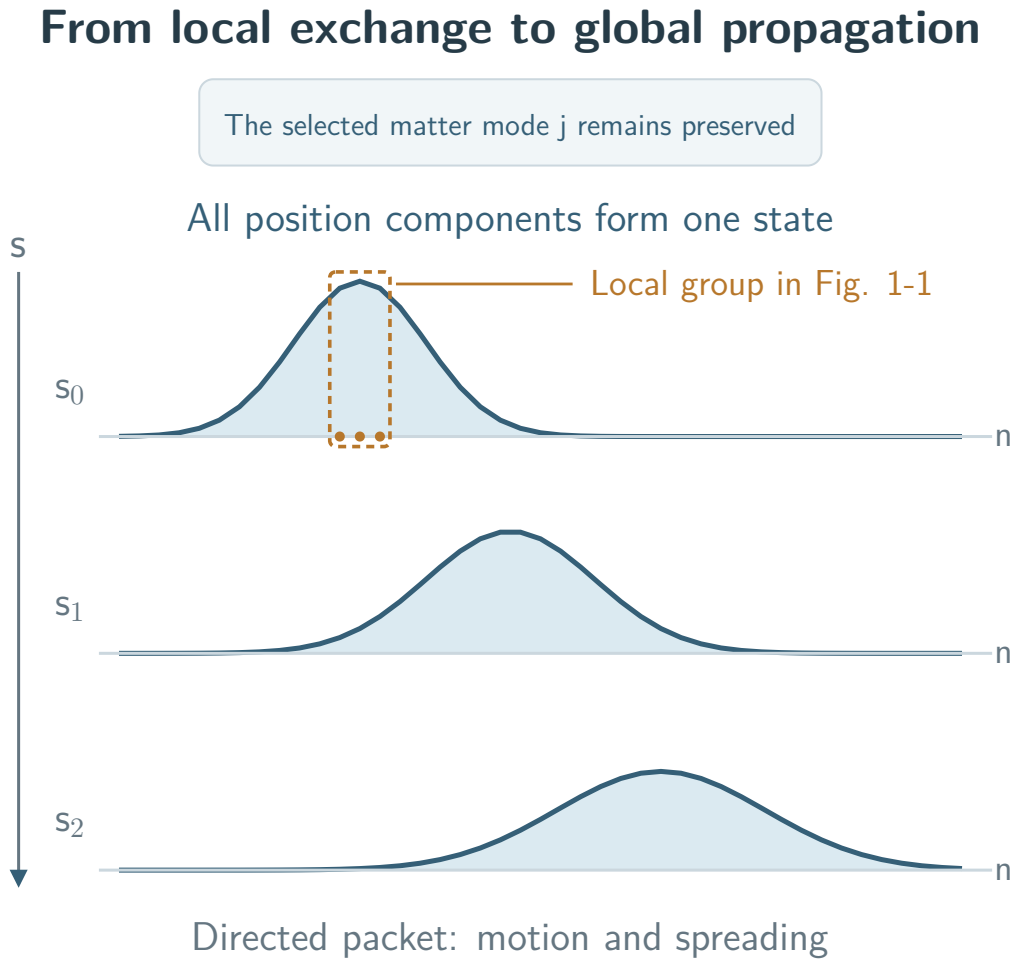


Figure 1-2. **From local exchange to global propagation.** The dashed box marks a group of neighboring positions of the kind shown in Figure 1-1. These local relations connect all position components. The curves illustrate the position distribution of a directed wave packet at three active times, while the selected matter mode j remains preserved. They represent distributions of the same preparation at different evolution times. These distributions can be compared using repeated trials terminated at different times, without inserting a position measurement at every step.

In uniform propagation, the global change can be described through the relative phase evolution of the propagation modes. This describes the same process as coherent amplitude exchange in the position representation. For a suitably narrow

spectral preparation confined to one propagation branch, the slope of the propagation spectrum determines the velocity of the wave-packet center, and its curvature controls dispersion. The internal mass action also affects this spectrum, linking the rest energy to the low-momentum inertial response. Local maintenance, global propagation, and the laws of motion are thereby connected; their specific relations are developed below.

The same matter also evolves jointly with surrounding matter, the background, and apparatus. Through interactions, surrounding matter and the background change its phase, orientation, and position amplitudes, and exchange energy and momentum with it, producing deflection, acceleration, or scattering. The complete output of one stage feeds into the next. The survival of the original matter, possible exits, and the correlations left behind are all tracked along this history. The time of motion is calibrated by all stages actually traversed.

The source, moving matter, surrounding matter, background, and apparatus all belong to one joint state. Distinctions preserved by their interactions can become internal records, which correspond to position and time events through actual readout and calibration. Maintenance, propagation, and exchange continue between the selected events. Continuous motion and its readable events together form the complete picture of motion.

Within this overall picture, coherent propagation, discrete detection, and macroscopic trajectories have their respective formation conditions. Coherent propagation along multiple paths jointly determines an interference distribution, while a specific detection produces a corresponding position record. A composite object can be approximately described by a continuous trajectory when its collective distribution is sufficiently concentrated and its spreading and environmental interactions satisfy the relevant conditions. The overall picture accommodates these phenomena, with their actual manifestation determined by the preparation, interactions, and recording procedure.

We address three questions:

1. **How does matter change position while preserving its own structure?** How do internal maintenance and exchange between neighboring positions jointly produce position currents and connect into the continuous propagation of the same matter?
2. **How does propagation leave identifiable records of motion?** How does apparatus preserve position distinctions, how does actual readout identify events, and how does the state following a record continue to propagate?
3. **How does the complete process correspond to physical phenomena of motion?** How do phase and interactions determine interference and scattering, and how do composite structure and environmental records support macroscopic trajectories?

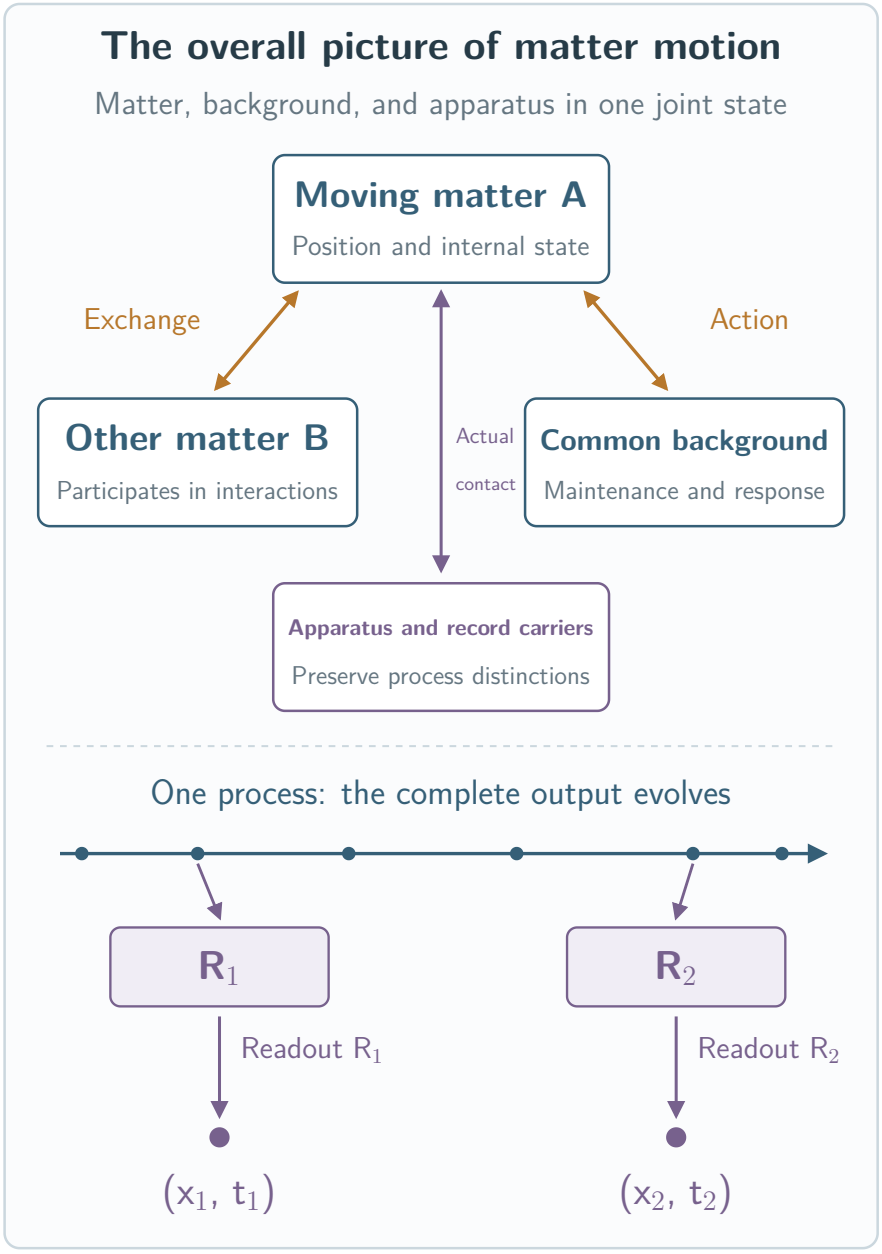


Figure 1-3. **The overall picture of matter motion.** The upper part shows the relations of moving matter to other matter, the common background, and apparatus; connecting lines denote the actual actions considered. The lower part unfolds the continuous history of the same process. The complete output continues to evolve, and preserved records R_1 and R_2 become events (x_1, t_1) and (x_2, t_2) through readout and calibration. Both records and readout are internal to the joint process. Only two records are selected here; motion between the events is determined by the actual dynamics.

§2 The Motion of Matter

2.1 Matter in Motion

An electron can have different position distributions while preserving its internal structure. It may be concentrated in one region, distributed over a larger region, or develop several coherent components during propagation. Internal maintenance allows the same electron to persist [2], while the position degree of freedom carries its spatial distribution [1].

Matter delivered by actual formation, together with the source, working auxiliary systems, and correlations still involved in its maintenance, constitutes a complete carrier [2]. Loading connects the formation output to continuing returns; formation records that still control contact are retained together with the current working state. Position changes occur in the matter state supported by this carrier.

Along one propagation direction, denote position in a finite position model by n, with $n = 0, \dots, L-1$ and periodic connections between neighbors. Let j label the selected matter mode of the complete internal action, and let $\chi_n(s)$ be the motion-orientation amplitude at position n. States that can vary independently within the carrier are called logical states. For a pure state, the state of motion is

$$\begin{aligned} \psi_j(s) &= \sum_n |j\rangle \otimes \chi_n(s) \otimes |n\rangle, \\ p_n(s) &= \chi_n^\dagger(s)\chi_n(s), \qquad \sum_n p_n(s) = 1. \end{aligned} \tag{2.1}$$

Here s is the active time during which maintenance and the position action take place, and normalization refers to the prepared surviving matter. The amplitude χ_n has two motion-orientation components, which may differ with position and remain correlated with it. All χ_n together constitute one matter state; their relative amplitudes and phases determine its spatial distribution and subsequent propagation.

The label n denotes a position in the propagation model, whose readout is determined by position detection [1]. The directions of the formation pattern and the counting addresses of the background each have their own definitions.

Comparing position statistics for identical preparations terminated at different times gives p_n at those times. Each actual detection also produces a corresponding record.

2.2 Maintenance and Propagation

After matter forms, the source, target, and auxiliary systems involved in the return continue to evolve jointly. Actual loading embeds the retained matter state into this joint space, and maintenance preserves the image of the loading map during the active stage [2]. Let the isometries J_0, J_1 describe loading in the two motion orientations, X_f the Hermitian operator for the actual contact connecting the two ends, and g_M its strength. Equal-strength, simultaneous two-arm maintenance applied to the complete loading [2] gives the internal generator

$$K_f = J_0^\dagger X_f J_1, \qquad h_{M,f} = g_M \begin{pmatrix} 0 & -iK_f \\ iK_f^\dagger & 0 \end{pmatrix}. \tag{2.2}$$

The internal blocks used here satisfy $K_f = K_f^\dagger$, so that, ordering the factors as internal mode and motion orientation, $h_{M,f} = M_f \otimes Y$, where $M_f = g_M K_f$ is the internal mass generator measured in active time and $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. The formation-source shares, actual paths, and return contacts enter the sustained phase through K_f; the selected mode satisfies $M_f|j\rangle = \mu_j|j\rangle$. The sign of the internal coefficient μ_j remains in the generator, while the rest half-gap is read per total elapsed time in Section 2.5.

Spatial propagation continues through the same loading. Let $\mathcal{J}$ be the complete isometry incorporating internal loading, position, and preparation of the common port, and take the logical displacement $S_x|n\rangle = |n+1\rangle$. The actual displacement S_{phys} and actual orientation operator Z_{phys} satisfy, on this carrier,

$$S_{\text{phys}}\mathcal{J} = \mathcal{J}S_x, \qquad Z_{\text{phys}}\mathcal{J} = \mathcal{J}Z_{\text{mov}}. \tag{2.3}$$

Here and below, identities on spectator factors are omitted when unambiguous. Position-independent internal loading directly accommodates displacement S_X of the matter address. For an up-type quark carrier whose contact depends on an accompanying resource position, actual loading maps matter displacement to $S_X S_R$, where R denotes the position of that contact resource [2]. Matter and the resource move together to the neighboring position, preserving the relative contact geometry used, and the old position is restored by the inverse return. A fixed internal matrix M_f describes uniform propagation preserving this contact geometry.

Two opposite position exchanges share a coherent port. Both working ends use the same maintenance action, and a definite relative phase connects neighboring amplitudes. Define

$$D_x = \frac{S_x^\dagger - S_x}{2i}, \tag{2.4}$$

and denote the position-exchange strength by g. Internal maintenance and position exchange jointly give

$$\boxed{h_f = M_f \otimes Y \otimes I_x + gI_f \otimes Z_{\text{mov}} \otimes D_x.} \tag{2.5}$$

The matrices Y and Z_{mov} act on the same two-dimensional motion-orientation space and satisfy $Y^2 = Z_{\text{mov}}^2 = I$ and $\{Y, Z_{\text{mov}}\} = 0$. We use $\hbar = 1$ below. With $\widehat{H}$ denoting the actual joint generator, the complete action satisfies

$$\boxed{\begin{aligned} \widehat{H}\mathcal{J} &= \mathcal{J}h_f, \\ e^{-is\widehat{H}}\mathcal{J}\psi_j(0) &= \mathcal{J}e^{-ish_f}\psi_j(0). \end{aligned}} \tag{2.6}$$

The joint state at any active time therefore remains $\mathcal{J}\psi_j(s)$. The first term generates internal evolution at each position; the second coherently connects orientation amplitudes at neighboring positions. The interval in which both terms act is called the active window. They generally do not commute: internal evolution affects subsequent position currents, while position exchange changes orientation–position correlations. The selected mode j is preserved, and all amplitudes χ_n evolve jointly within the same carrier.

2.3 Position Amplitudes and Position Currents

At position n, the amplitude obeys

$$\boxed{i\frac{d\chi_n}{ds} = \mu_j Y \chi_n + \frac{g}{2i} Z_{\mathrm{mov}}(\chi_{n+1} - \chi_{n-1}).} \tag{2.7}$$

The first term on the right comes from the internal action on the component already at that position; the second comes from coherent contributions from its left and right neighbors. After a short active interval Δs, the amplitude at the same position becomes

$$\begin{aligned} \chi_n(s + \Delta s) = \chi_n(s) & \\ & - i\mu_j Y \chi_n(s)\, \Delta s \\ & + \frac{g\, \Delta s}{2} Z_{\mathrm{mov}} \chi_{n-1}(s) \\ & - \frac{g\, \Delta s}{2} Z_{\mathrm{mov}} \chi_{n+1}(s) + O(\Delta s^2). \end{aligned} \tag{2.8}$$

The original amplitude, internal change, and neighboring contributions combine coherently in the same output. Their relative phases determine enhancement or cancellation, and the motion orientation determines how each component enters the exchange.

Position probabilities change with these amplitudes. Define the directed position current from n to $n+1$ by

$$\mathcal{I}_{n+1/2}(s) = g\, \mathrm{Re}\left[\chi_n^\dagger(s) Z_{\mathrm{mov}} \chi_{n+1}(s)\right], \tag{2.9}$$

Using the amplitude equation and $p_n = \chi_n^\dagger \chi_n$, first differentiate the probability:

$$\begin{aligned} \frac{dp_n}{ds} &= 2\, \mathrm{Re}\left(\chi_n^\dagger \frac{d\chi_n}{ds} \right) \\ &= g\, \mathrm{Re}(\chi_n^\dagger Z_{\mathrm{mov}} \chi_{n-1}) \\ &\quad - g\, \mathrm{Re}(\chi_n^\dagger Z_{\mathrm{mov}} \chi_{n+1}). \end{aligned} \tag{2.10}$$

The internal term contributes zero because $\chi_n^\dagger Y \chi_n$ is real. Hermiticity of Z_{mov} then allows the two positions in the first term to be exchanged, giving

$$\boxed{\frac{dp_n}{ds} = \mathcal{I}_{n-1/2} - \mathcal{I}_{n+1/2}.} \tag{2.11}$$

The probability increase at a position equals the difference between inflow and outflow; the sign of the position current gives the actual direction of flow. Summing over all positions cancels contributions from neighboring edges, preserving total probability during the active stage. These local balances jointly realize the spatial redistribution of matter.

The internal mass term still affects subsequent position currents through the motion orientation and the relative relations of neighboring amplitudes. Knowing the probabilities alone is therefore insufficient to determine the distribution at the next instant.

If the preparation at some instant has nonzero amplitude at only one position, neighboring positions acquire first-order amplitudes over a short interval, and their

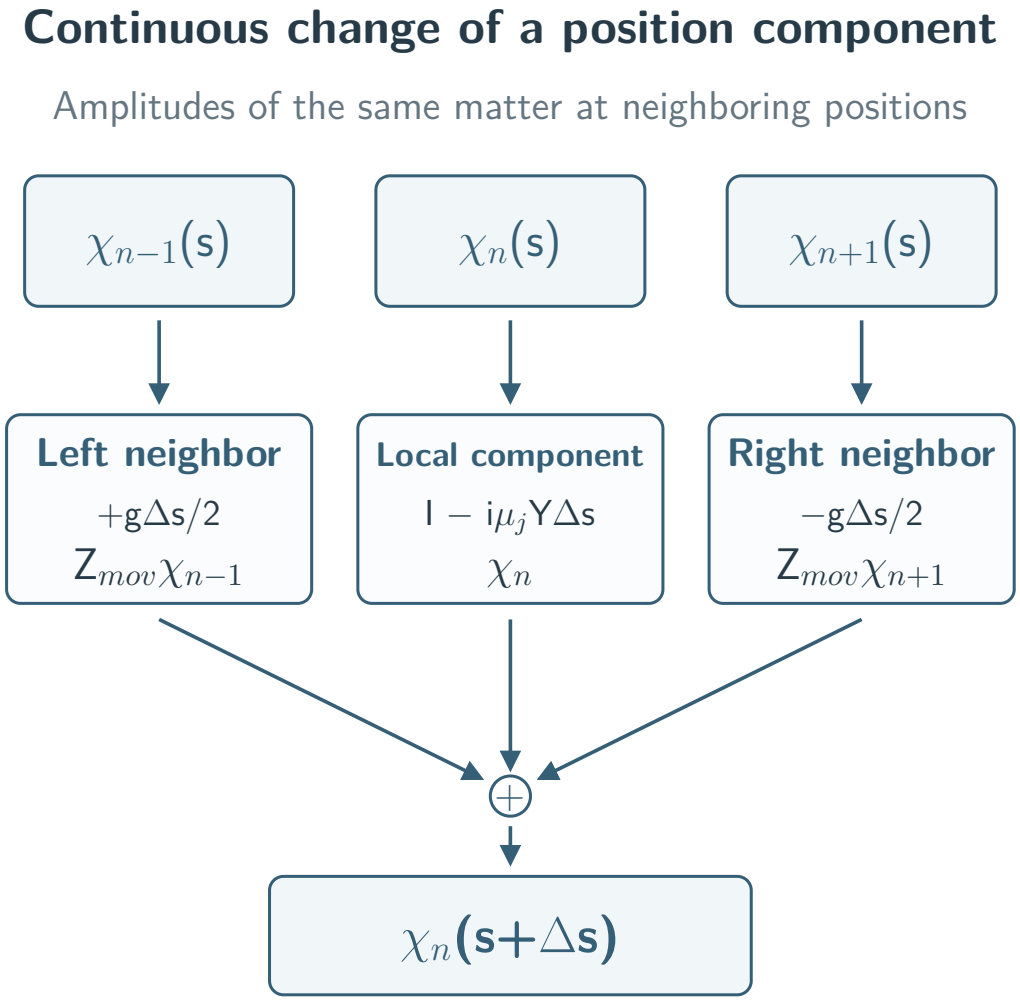


Figure 2-1. **Continuous change of a position component.** The upper part shows the orientation amplitudes of the same matter at three neighboring positions. The original component at position n undergoes internal evolution, while the left and right neighbors simultaneously contribute amplitudes with relative signs, together forming $\chi_n(s + \Delta s)$ below. Arrows indicate amplitude contributions over the same short interval; the output still belongs to matter mode j. The diagram corresponds to the first-order expansion in the text; the same generator gives the complete evolution.

probabilities begin to increase at second order. As these components evolve, further exchanges between neighbors continue. Spatial propagation thus develops across the distribution; where it concentrates depends on the amplitudes, phases, and orientations in the complete preparation.

2.4 Wave-Packet Propagation

Local position currents jointly change the wave-packet distribution. In the propagation-mode representation, this change is described by the relative phase evolution of the components.

For uniform action, use the spatial phase convention $e^{ikn\ell_x}$, where ℓ_x is the calibrated spacing of the selected uniform position graph [2]. Within this regime, take $x_n = n\ell_x$; the allowed wave numbers of the finite ring are $k = 2\pi r/(L\ell_x)$. The metric of the underlying fine positions and its correspondence to this coordinate remain subject to the established calibration; they cannot be identified merely from neighboring labels. Then $D_x(k) = \sin(k\ell_x)$, and the same action becomes

$$h_f(k) = M_f \otimes Y + gI_f \otimes Z_{\mathrm{mov}} \sin(k\ell_x). \tag{2.12}$$

Fix a matter mode j and select one propagation branch. Let $u_j(k)$ be its normalized motion-orientation eigenvector and $\epsilon_j(k)$ the eigenvalue of the active generator. The wave packet can be written as

$$\begin{aligned} \chi_n(s) &= \frac{1}{\sqrt{L}} \sum_k a(k) u_j(k) e^{ikn\ell_x - i\epsilon_j(k)s}, \\ \sum_k |a(k)|^2 &= 1. \end{aligned} \tag{2.13}$$

The magnitudes and phases of the initial coefficients $a(k)$ determine the preparation. During evolution, each propagation component accumulates phase at its own frequency, and their relative changes alter the global position distribution. Since $u_j(k)$ generally depends on wave number, the motion orientations at different positions are also jointly determined by this superposition. A preparation containing several propagation branches retains all of them in the expansion.

The effect of internal maintenance on spatial propagation can be read directly from the spectrum of this joint action. The anticommutation relation of the two orientation matrices gives [2]

$$\epsilon_{j,\pm}(k) = \pm\sqrt{\mu_j^2 + g^2 \sin^2(k\ell_x)}. \tag{2.14}$$

For a narrow spectral wave packet confined to one propagation branch, within a regime admitting a continuous wave-number description and before position wraparound occurs, the nearly linear phase over the occupied interval moves the concentrated region. The group velocity measured in active time is $v_s = \partial_k \epsilon_j(k)$. Spectral curvature over the same interval gives different components different velocities, so the distribution may spread as it moves. Figure 2-2 shows an example under the same generator.

Different preparations produce different motion. When the initial state and action jointly preserve the relevant spatial symmetry, the distribution may spread

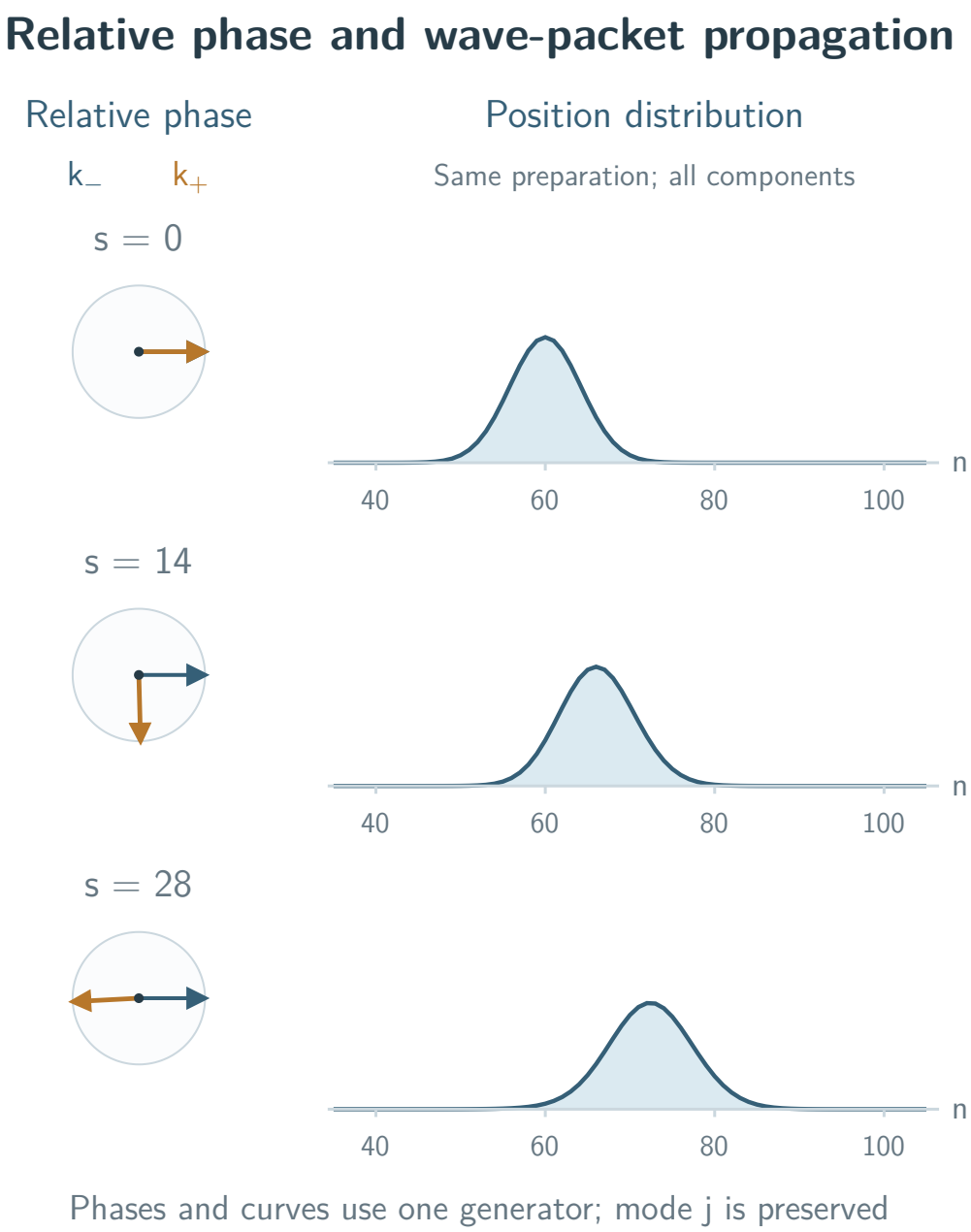


Figure 2-2. **Relative phase and wave-packet propagation.** On the left, two representative wave numbers in the packet illustrate the relative phase change, using the lower-wave-number component as the phase reference. On the right, all wave-number components are used to calculate the position distribution of the same packet. The three times use the same initial state and the generator combining maintenance and displacement; the packet moves and spreads. The parameters are $L = 257$, $\mu_j = 0.6$, and $g = \ell_x = 1$; the initial positive-energy-branch wave-number distribution is centered at $k_0 = 0.35$ with standard deviation 0.12. These parameters illustrate the propagation structure and do not provide a numerical calibration for the electron.

in both directions. A directed preparation may produce a moving concentrated region, and several propagation components may form a split wave packet.

A single complete eigenmode accumulating only a common phase has an unchanged position probability. A localized wave packet changes through the relative evolution of its propagation components. Position-probability change and position current describe different physical quantities: when inflow equals outflow, the local probability may remain unchanged despite a nonzero current.

2.5 Continuous Motion

The complete state from one stage of evolution becomes the input to the next, so the continuous history retains amplitudes, phases, and correlations together.

This history also includes the return, thermal processing, and possible exits that actually occur. Carrier preservation during the active stage allows surviving matter to continue propagating; the complete process also keeps track of the share of the original matter remaining in this branch. For a scalar return process [2], take a normalized logical state ρ_0 after loading, an active duration t_H per cycle, survival probability $0 < q_T \leq 1$, and $U = e^{-it_H h_f}$. Under the record-factorization conditions of this realization, the surviving component after N consecutive cycles is

$$\rho_{\text{live}}^{(N)} = q_T^N \mathcal{J} U^N \rho_0 U^{\dagger N} \mathcal{J}^\dagger. \tag{2.15}$$

Here q_T^N gives the share of the original matter that survives, while the normalized state $U^N \rho_0 U^{\dagger N}$ gives the position and internal state of the survivors. The loss of surviving share and the spatial propagation of survivors are read separately along the same history. Exit components enter the actual subsequent process; excitations formed afterward have their own formation histories. The complete output retains these destinations and the corresponding records. More general background correlations remain part of their actual joint evolution.

Elapsed time is also accumulated along this history. If each complete cycle has fixed duration T_{cyc}, then N cycles correspond to

$$s_N = N t_H, \qquad t_N = N T_{\text{cyc}}, \qquad f_{\text{act}} = \frac{t_H}{T_{\text{cyc}}}. \tag{2.16}$$

The time s_N measures the active duration of maintenance and position exchange; t_N also includes the return, repair, control, and waiting actually performed each cycle. Initial formation and initial loading contribute to startup time; loading and inverse operations repeated each cycle enter T_{cyc}, and final readout adds its actual duration separately [2]. Many propagation stages may separate successive records, and the number of selected records is tracked separately from the number of cycles actually traversed. Position change and phase accumulation are calibrated along the same elapsed duration, yielding comparable velocities and frequencies.

Under these fixed-period and survival conditions, the propagation phase at cycle endpoints satisfies $\epsilon_j(k) s_N = E_j(k) t_N$, where $E_j(k) = f_{\text{act}} \epsilon_j(k)$. This gives the propagation energy read with respect to the total elapsed duration. In an effective continuous description between cycle endpoints, the corresponding center velocity is $v_g = \partial_k E_j(k)$. Motion resolved within a single cycle must still be calculated along the actual active intervals and other actions; a cycle average cannot replace intracycle evolution. Wave packets and velocities calculated from E_j below use this

effective description at cycle endpoints or on the cycle timescale. For a massive mode $\mu_j \neq 0$, use the long-wavelength momentum $P = k$ near the origin of the positive-energy branch. Denote its rest energy by E_{0j}, equal to the half-gap of the selected symmetric propagation spectrum. Then

$$\begin{aligned} E_+(P) &= E_{0j} + \frac{P^2}{2M_{\mathrm{in},j}} + O(P^4), \\ E_{0j} &= f_{\mathrm{act}}|\mu_j|. \end{aligned} \tag{2.17}$$

The energy E_{0j} is already calibrated per total elapsed time; formation yield and resource-supply probabilities determine the prepared share of the branch. The inertial mass $M_{\mathrm{in},j}$ is determined by the curvature at the origin. Writing $c_0 = f_{\mathrm{act}} g \ell_x$ for the velocity scale of the massless branch near the origin, the same spectrum gives

$$\boxed{\begin{aligned} \frac{1}{M_{\mathrm{in},j}} &= \frac{f_{\mathrm{act}} g^2 \ell_x^2}{|\mu_j|}, \\ M_{\mathrm{in},j} c_0^2 &= E_{0j}. \end{aligned}} \tag{2.18}$$

Thus, at low momentum, the rate of change of velocity with momentum is $1/M_{\mathrm{in},j}$. The rest energy determined by internal maintenance also enters the response of motion; the same curvature affects the spreading of a wave packet with finite spectral width. In uniform propagation, occupation at each wave number is preserved, so the mean group velocity of a single-branch wave packet is preserved even though its position distribution may change. The full lattice spectrum retains its sine form, and the long-wavelength relations apply within their corresponding regime.

The photon's rest-mass term vanishes, while its position action remains [2]. Its velocity and dispersion are determined by the actual position spectrum and the wave-number interval used. When apparatus interacts with matter and preserves a position distinction, propagation becomes connected to readable events.

§3 Records of Motion

During exchanges between matter and the environment or an apparatus, different propagation processes can leave distinctions in other degrees of freedom. The preservation and readout of these distinctions by apparatus make motion manifest as comparable position and time events.

3.1 Criteria for a Record

Three conditions jointly determine whether a distinction left during propagation constitutes a complete record [1,2].

First, the distinction must be identifiable. Different process classes must correspond to states that can be distinguished without error by the actual pointer, that is, to different orthogonal pointer subspaces. The pointer may be carried by apparatus or other environmental degrees of freedom. The content of the record is determined by the correspondence established by the actual action. For example, if the pointer distinguishes spatial regions traversed by matter, it preserves a position distinction.

Second, the distinction must actually remain. The criterion applies to the complete output, including retrieval and pattern reconstruction: a nonzero component carrying the distinction must still exist after these actions are complete. If an auxiliary system occupies different states during different processes but ultimately returns to a common preparation and decouples from the selected process, that auxiliary retains no such record. Whether other outputs preserve the distinction must still be judged from their actual states.

Finally, the distinction must persist. From completion of the record until its readout, subsequent action must preserve the relevant pointer label so that the earlier distinction remains identifiable [3]. Preventing the record-carrying degree of freedom from participating in later exchanges can protect it from retrieval.

If some part of the output still overlaps, that part alone does not justify setting the corresponding matter cross term to zero; all other correlated outputs must be included when determining whether coherence remains. Matter can also continue to propagate without leaving the selected record, and its coherent evolution still affects the statistics of later events.

3.2 Position Records

Retain the prepared matter mode j of Chapter 2, and suppose that the wave packet is still distributed over several positions when it reaches the detection region. The spatial range resolved by the apparatus is determined by its preparation and action. Partition the finite position graph into disjoint regions A. On the logical motion space of Chapter 2, the corresponding projections are

$$\begin{aligned} Q_A &= I_f \otimes I_{\mathrm{mov}} \otimes \sum_{n \in A} |n\rangle\langle n|, \\ \sum_A Q_A &= I. \end{aligned} \tag{3.1}$$

Here n labels a position in the propagation model, and a region A may contain one or more position components. The projection Q_A acts as the identity on

the matter mode and motion orientation, identifying the spatial distribution of the same matter. The loading used in Chapter 2 preserves the matter position address: position-independent internal loading commutes with these projections, and the accompanying-position loading for the up-type carrier also preserves the matter address X. Let $\widehat{Q}_A$ project, in the complete space, onto matter located in region A, acting as the identity on the remaining factors. Then

$$\widehat{Q}_A \mathcal{J} = \mathcal{J} Q_A. \tag{3.2}$$

Resolution by matter position therefore retains the accompanying resources and correlations within the corresponding carrier. Outputs outside the carrier are retained separately by the complete instrument.

Detection correlates distinguishable components with orthogonal pointers [1]. As an ideal realization of a position record, take a finite detection interval of duration t_D, during which position exchange of the complete carrier is paused and internal maintenance continues. Denote the internal evolution during this interval by U_m, which commutes with every Q_A. Let $|g_D\rangle$ be the pointer preparation and let orthogonal states $|r_A\rangle$ preserve the region labels. The position-controlled action $\sum_A \widehat{Q}_A \otimes V_A$, with pointer unitaries V_A satisfying $V_A|g_D\rangle = |r_A\rangle$, therefore gives on this preparation

$$\boxed{\begin{aligned} \mathbb{U}_D(\mathcal{J}\psi_j \otimes g_D) &= \sum_A \mathcal{J} Q_A U_m \psi_j \otimes r_A, \\ \langle r_B | r_A \rangle &= \delta_{AB}. \end{aligned}} \tag{3.3}$$

Each position component becomes correlated with the corresponding pointer; components in the same region share a label. When a broader region is resolved, position coherence within that region may remain. Figure 3-1 shows this joint output.

If this pointer distinction survives subsequent retrieval and storage, detection satisfies the criteria of Section 3.1. If the apparatus distinguishes the two regions only partially, the corresponding cross coherence must be retained according to the actual overlap of the final environmental states.

A channel label in maintenance need not resolve motion position and cannot simply be renamed a position reading. If position exchange continues during detection, propagation generally does not commute with the position projections; the complete output must be calculated using the joint generator for that interval.

3.3 Event Readings

After a position record is left, actual readout identifies a pointer outcome and assigns a spatial reading using the apparatus calibration. An outcome identified by an actual readout is called an event [1]. A region label A determines the corresponding spatial resolution range. If position is reported by the region center x_A, the region width also specifies the resolution scale of the reading.

Let $\Omega^{\mathrm{out}}(\rho)$ be the joint state after an actual detection is complete, and let Π_A^{rec} be the pointer projection preserving outcome A, with $\widehat{\Pi}_A = I \otimes \Pi_A^{\mathrm{rec}}$ on the joint space. Taking a partial trace over detection degrees of freedom whose action is complete and which will not recouple gives the conditional action and probability for the outcome:

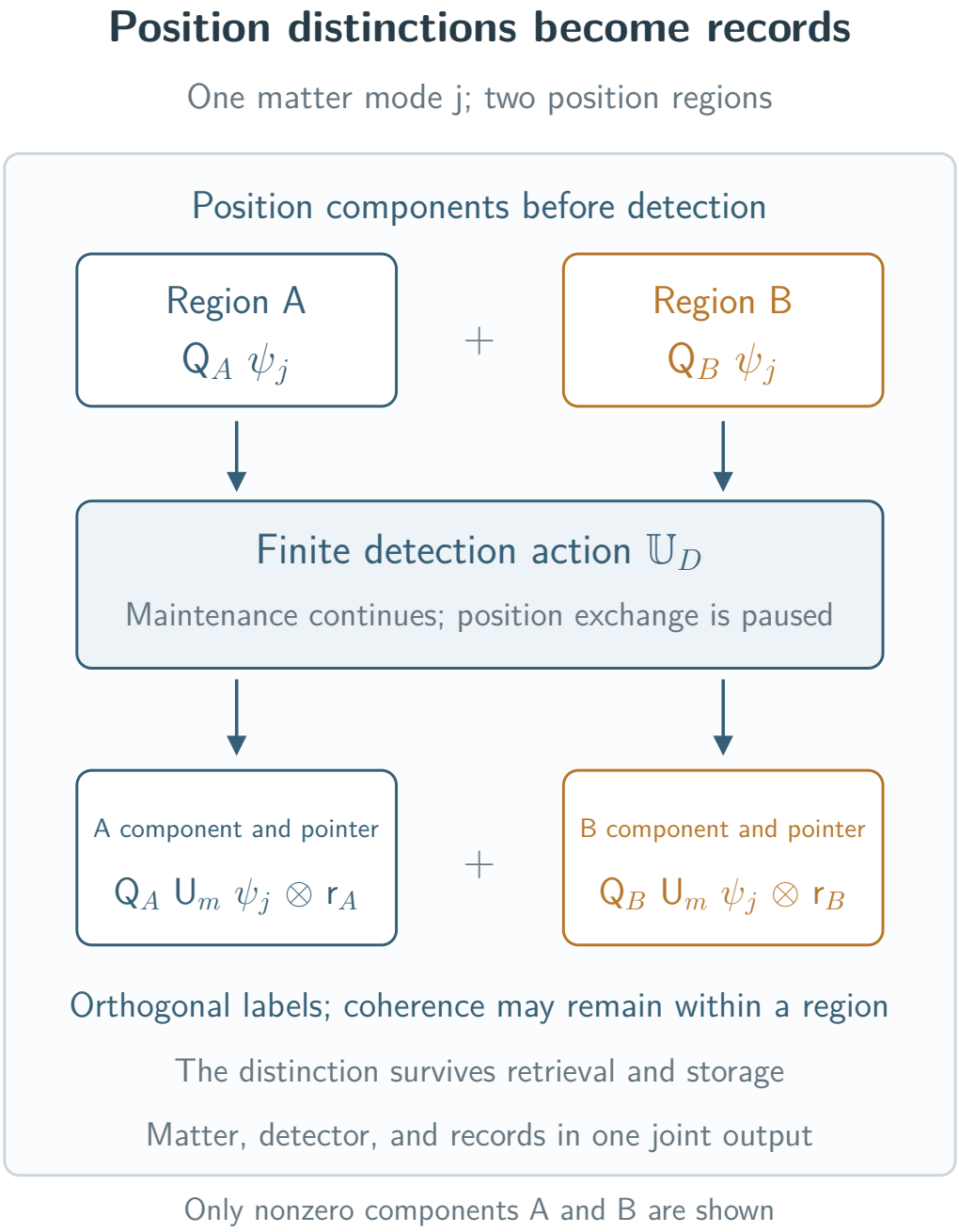


Figure 3-1. Position distinctions become records during propagation. The nonzero components in regions A and B belong to the same matter state. Finite detection correlates them with distinguishable pointer states. The distinction must remain in the output after subsequent retrieval and be preserved during readout. The diagram uses an ideal readout with position exchange paused; internal maintenance and detection duration are retained. Regions not shown and no-response outcomes are included in the complete instrument.

$$\boxed{\begin{aligned}\mathcal{I}_A(\rho) &= \mathrm{Tr}_D\left[\widehat{\Pi}_A \Omega^{\mathrm{out}}(\rho)\widehat{\Pi}_A\right], \\ p_A &= \mathrm{Tr}\,\mathcal{I}_A(\rho).\end{aligned}} \tag{3.4}$$

Here ρ includes the matter, background, and auxiliary systems that still participate in subsequent action; D denotes detection degrees of freedom that no longer couple to the subsequent system. The conditional actions for all outcomes form a quantum instrument [4,5]. Exit, absorption, and no-response outcomes must be included in the complete statistics [1]. For the surviving carrier, identify the state with the logical space through $\mathcal{J}$, retaining the notation ρ in this representation. Ideal position readout then becomes

$$\begin{aligned}\mathcal{I}_A(\rho) &= Q_A U_m \rho U_m^\dagger Q_A, \\ p_A &= \mathrm{Tr}(Q_A \rho).\end{aligned} \tag{3.5}$$

This connects the position distribution of Chapter 2 to position events. For a pure state ψ_j, the probability equals the sum of p_n over the region. Repeating the same preparation and detecting at the same stage of operation gives the corresponding position statistics. In a single run, position records enter the history according to the outcomes actually identified.

The elapsed time to completion of a record includes the detection duration in addition to the propagation duration calculated in Chapter 2. If loaded matter first undergoes N cycles of duration T_{cyc} and then one detection of duration t_D, the record is complete at

$$t_{\mathrm{read}} = t_0 + NT_{\mathrm{cyc}} + t_D. \tag{3.6}$$

Here t_0 is the completion time of loading, preceded by the duration of initial preparation. If the actual apparatus resolves the response time within the detection interval, the corresponding time pointer should be used. If only a terminal outcome is available, the reading retains the detection time range.

The time reference also belongs to the joint process. Coherence between two components can establish a phase reading [2]. Under fixed-period and appropriate survival conditions, if their active eigenfrequencies are ϵ_a and ϵ_b, their readable relative frequency is $\omega_{ab} = f_{\mathrm{act}}(\epsilon_a - \epsilon_b)$. Over an interval with nonzero reference coherence, calibrated frequency, and continuously unwrapped phase, a phase change corresponds to

$$\Delta t = -\frac{\Delta\widetilde{\varphi}_{ab}}{\omega_{ab}}, \qquad \omega_{ab} \neq 0. \tag{3.7}$$

The real and imaginary parts of the coherence matrix element are read in the respective measurement settings, while the surviving share determines the statistical weight of available samples. If detection changes the actual operating schedule of the timing reference, its frequency must be recalibrated using the complete process including detection. The fixed-period relation applies where the original schedule is satisfied. A phase clock requires an actual reference and coherent readout. A further interaction with a reference field can convert the coherence phase into a transition probability; Ramsey's separated oscillatory fields provide a concrete realization [6]. Comparable spatial and temporal events arise when the apparatus correlates this calibrated time information with position labels.

3.4 Successive Records

After a detection is complete, subsequent motion continues from the state it leaves. For an outcome A of nonzero probability, the conditional state is

$$\rho_A = \frac{\mathcal{I}_A(\rho)}{p_A}. \tag{3.8}$$

In the ideal detection above that preserves the matter carrier, mode j remains preserved, and the position distribution is restricted to the resolved region. Coherence within the region and motion orientation are still carried by ρ_A. The position action switched on afterward acts on this state, which also determines the new currents between positions. If the data are not separated according to pointer outcome, the post-detection state is the sum of all outcome components. Ideal readout removes coherence between different regions in the reduced matter state and therefore affects subsequent propagation statistics.

Let $\mathcal{G}_1$ describe actual propagation from completion of preparation to the first detection, and $\mathcal{G}_2$ propagation from completion of the first detection to the second. Let $\mathcal{I}_A^{(1)}$ and $\mathcal{I}_B^{(2)}$ be the conditional actions for the respective outcomes in the two instruments. Feeding the first output into the second stage gives

$$\boxed{\begin{aligned} \rho_{AB}^{\mathrm{un}} &= \mathcal{I}_B^{(2)} \circ \mathcal{G}_2 \circ \mathcal{I}_A^{(1)} \circ \mathcal{G}_1(\rho_0), \\ p(A,B) &= \operatorname{Tr} \rho_{AB}^{\mathrm{un}}. \end{aligned}} \tag{3.9}$$

Here ρ_{AB}^{un} is the unnormalized output for this record history, and ρ_0 is the initial state after loading. Each stage acts on a sufficiently large joint space: previous background and auxiliary systems that will participate again remain included. Degrees of freedom that no longer couple to the subsequent system can be traced out after the selected record labels are retained. With independent fresh detection preparations, this is sequential composition of quantum instruments [1]; more general correlated preparations are included directly in the joint state.

Correlations between two events are determined by the state following the first record. Omitting the first detection changes the input to the second stage; ignoring its outcome merely sums the outputs of a detection that has already occurred. Position distributions obtained in independent trials terminated at different times therefore generally yield different subsequent statistics from successive position records in a single run. The multiple-time wave-packet plots in Chapter 2 describe the former comparison; successive events are calculated from the actual history above.

As records accumulate, calibrated events (x_A, t_A) and (x_B, t_B) give a displacement and time interval. Within the same position-coordinate range, $(x_B - x_A)/(t_B - t_A)$ gives the average velocity between events, with resolution limited jointly by the spatial and temporal ranges. This reading does not supply all unrecorded positions between detections; the intervening state is determined by the retained dynamics and correlations.

An event sequence can be approximated by a smooth trajectory only when the conditional distributions associated with successive records remain concentrated, and spreading and detection-induced changes between records are controlled on the scale considered. Finer position resolution also changes the momentum composition of the packet, so more frequent recording alone does not guarantee this approxima-

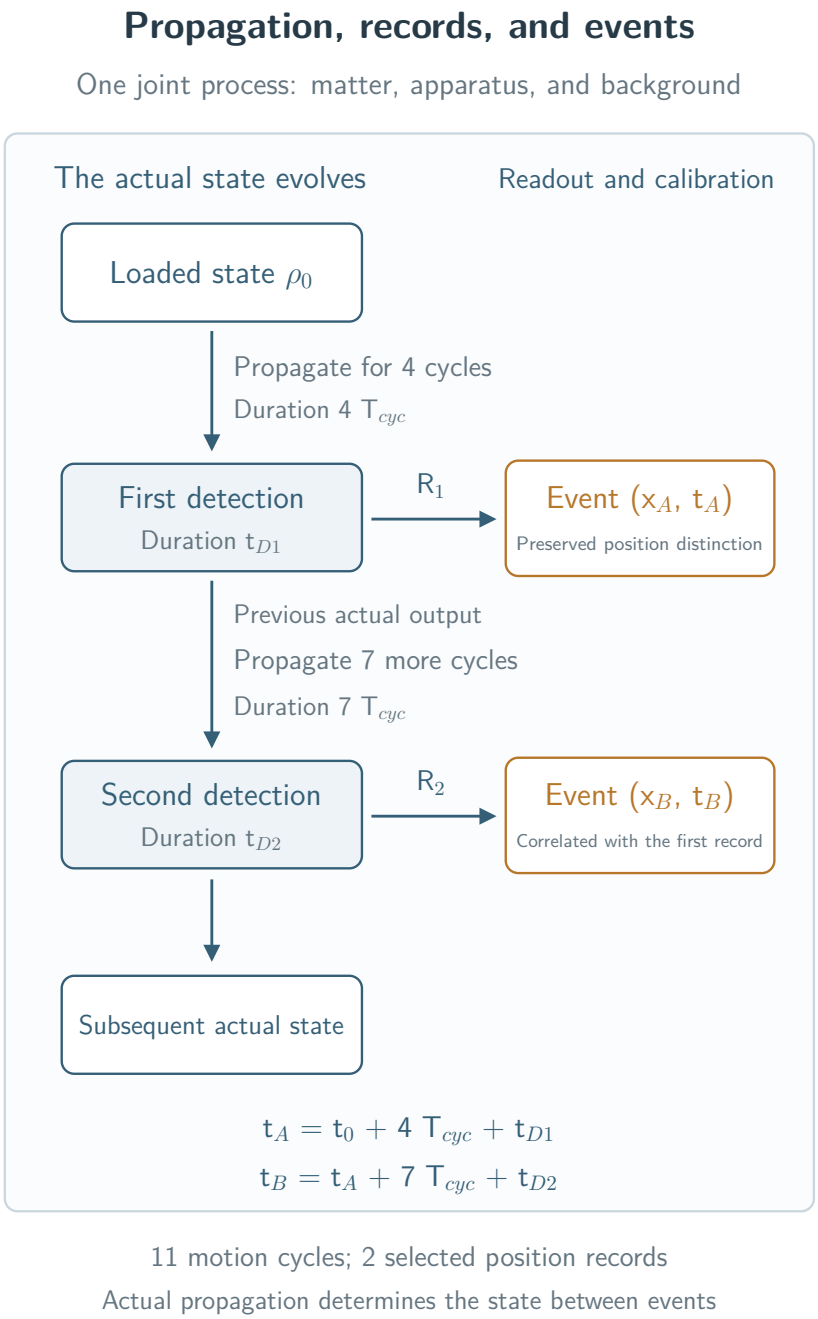


Figure 3-2. A continuous history of propagation, records, and events. Matter and correlations still participating in the action continue to evolve vertically; records left by two finite detections correspond to events along the readout arrows. The actual output of the first detection enters the next propagation stage. In this example, detection occurs after four cycles and again after seven more, for a total of eleven motion cycles and two selected position records. Both detection durations enter the event times. No point-by-point trajectory is presupposed between events.

tion. The concentration of macroscopic objects, environmental action, and recording conditions are considered together below in the specific phenomena of motion.

Continued tracking also requires the original matter to survive between actions. An absorptive detector may leave a terminal record while ending the subsequent propagation of the original matter; excitations formed after an exit have their own formation histories. Whether a record preserves the original matter, changes its momentum, or permits further detection is determined by the actual output.

§4 Conservation Laws of Motion

An encounter between matter and a detector can both leave a record and alter subsequent motion. Position detection in Chapter 3 can change the matter's position distribution and coherence, and may also change its momentum distribution. The detector, environment, and control degrees of freedom jointly implement the action; changes in matter itself must be compared within their complete exchange. Conservation relations are preserved by the actual dynamics and appear as input–output balances in records that resolve the corresponding exchanged quantities.

4.1 Conserved Quantities in Propagation

Conserved quantities in uniform propagation are determined by commutation relations with the generator.

Retain matter mode j and wave number k on the finite position ring. Let P_j be the matter-mode projection and P_k the wave-number projection including all motion orientations. The generator of Chapter 2 satisfies

$$[h_f, \mathsf{P}_j] = 0, \qquad [h_f, \mathsf{P}_k] = 0. \tag{4.1}$$

The selected matter therefore retains its identity during this propagation, and the total share at each signed wave number is separately preserved. Motion orientation can change within a given k component. It helps determine the position current, but an orientation flip does not replace wave number k by $-k$. For a single-branch wave packet, preserved wave-number weights and the fixed propagation spectrum preserve the mean group velocity, while relative phases still make the position distribution change in time.

The finite-difference momentum operator on the same position ring is [2]

$$\begin{aligned} \Pi_x &= I_f \otimes I_{\mathrm{mov}} \otimes D_x/\ell_x, \\ \Pi_x(k) &= \frac{\sin(k\ell_x)}{\ell_x}. \end{aligned} \tag{4.2}$$

It commutes with uniform h_f, so its distribution and expectation are preserved during this propagation. Only in the long-wavelength regime, where $\Pi_x(k)$ approaches k, does it approximate the continuous momentum used in Chapter 2. Wave number, finite-difference momentum, and motion orientation have distinct roles in the finite position model; actual readings are compared using their respective calibrations.

Propagation energy is likewise preserved by the fixed action. In the fixed-period, scalar-survival realization of Chapter 2, normalized propagation compared at complete-cycle endpoints has generator $H_{\mathrm{prop}} = f_{\mathrm{act}} h_f$ and spectrum $E_j(k)$. The weights of its energy components are preserved; the position distribution changes through relative phases between components. When survival probability decreases, these properties describe the original matter that survives. The share carried away by exit outputs remains part of the complete process.

This propagation energy is determined by active phase and actual elapsed duration. Energy at thermal-exchange endpoints is described by the shell-energy operator H_K [1]; the two have different roles in the construction. Applying a propagation energy difference to actual emission or absorption requires compatibility between the

joint generator and environmental energy [2]. The actual relay exchange [2] transfers amplitudes between resonant levels with matched energy and charge changes, restoring the mediator to its common preparation. This process connects propagation carriers to comparable exchanged quantities; Chapter 5 uses it to construct an entangled pair.

4.2 Conservation in Exchanges

The participants in an exchange include auxiliary systems that will still return, the environment that actually receives the exchanged quantity, and the references used in preparation and control. Retaining these degrees of freedom in the joint process is necessary to determine the change on each side.

The ideal position detection of Chapter 3 gives a direct example. Set $\sigma = U_m \rho U_m^\dagger$. Summing all region outcomes gives $\sigma' = \sum_A Q_A \sigma Q_A$. By cyclicity of the trace, the change in matter momentum across detection is

$$\begin{aligned} \Delta\langle \Pi_x \rangle &= \mathrm{Tr}[\Pi_x(\sigma' - \sigma)] \\ &= \mathrm{Tr}\left[\left(\sum_A Q_A \Pi_x Q_A - \Pi_x\right)\sigma\right]. \end{aligned} \tag{4.3}$$

Region projections generally do not commute with the neighboring-position finite-difference momentum, so matter momentum can change after summing the recorded outcomes. Even when the mean change happens to vanish for a particular preparation, its momentum distribution may still change. The change is produced by the actual detection action; how the apparatus accommodates the balance must be checked in the complete joint action.

Joint energy conservation over a complete cycle is determined by commutation of the actual action with total energy [1]. Let H_S be the energy of matter and the retained system, and $H_\mathcal{E}$ the energy of all exchange auxiliary systems. At input and output endpoints where interaction is off, total energy is their sum. For a complete action $\mathbb{U}$ satisfying this conservation relation,

$$\boxed{\begin{aligned} [\mathbb{U}, H_S \otimes I + I \otimes H_\mathcal{E}] = 0, \\ \Delta\langle H_S \rangle = -\Delta\langle H_\mathcal{E} \rangle. \end{aligned}} \tag{4.4}$$

Energy gained by matter comes from the environment, and energy lost by matter enters other outputs. When an auxiliary system is completely retrieved, its entire input and output are compared. Exit and absorption components also enter the total balance. While interaction remains on, joint energy includes the interaction term; the energies associated with the uncoupled Hamiltonians on the two sides cannot simply be added.

The motion action also involves finite switching on and off. In a realization whose control is described by a prescribed schedule, control may do work, and the energy required to switch the interaction on and off must be included [7]. Let H_0 be the bare energy, H_{on} the actual generator of the active window, and $D = H_{\mathrm{on}} - H_0$. This fixed generator connects states ρ and ρ' before and after the active stage. Taking energy supplied by control to the system as positive, rectangular switching on and off does work $W_{\mathrm{on}} = \mathrm{Tr}(D\rho)$ and $W_{\mathrm{off}} = -\mathrm{Tr}(D\rho')$, respectively [2], satisfying

$$W_{\mathrm{on}} + W_{\mathrm{off}} = \mathrm{Tr}[H_0(\rho' - \rho)]. \tag{4.5}$$

If the controller is included in a closed joint system, its energy change accounts for control work. Preserving a matter mode and preserving its bare energy are different conditions; the former alone does not eliminate these exchanges.

Momentum transfer must also be calculated from the actual joint action. The commutator of a nonuniform maintenance term with Π_x determines the impulse on matter [2]; this contribution vanishes in a uniform background. If a complete exchange preserves the additive total momentum $P_{\mathrm{tot}} = P_S \otimes I + I \otimes P_{\mathcal{E}}$, then

$$[\mathbb{U}, P_{\mathrm{tot}}] = 0 \quad \Longrightarrow \quad \Delta\langle P_S \rangle = -\Delta\langle P_{\mathcal{E}} \rangle. \tag{4.6}$$

Recoil in scattering and impulse borne by apparatus supports belong to this joint comparison. Treating apparatus as fixed at a location approximates its motion; the momentum transferred when it acts on matter must still be borne by the apparatus and its supporting environment. On a finite position ring, joint discrete translation symmetry first preserves the eigenphase of total translation, that is, total wave number modulo a reciprocal lattice vector. Whether the sum of the finite-difference momenta of the separate sides is preserved requires the corresponding joint commutation relation. Addition of continuous long-wavelength momenta applies in the appropriate effective regime.

Changes of matter species are constrained by joint unit-group charge [1]. A multiplicative character of the unit group is a group homomorphism to the unit circle. Let χ_S and $\chi_{\mathcal{E}}$ be definite characters of input and output components. If the actual action preserves the total unit-group action, every nonzero transition component satisfies

$$\chi_S^{\mathrm{out}}(u)\chi_{\mathcal{E}}^{\mathrm{out}}(u) = \chi_S^{\mathrm{in}}(u)\chi_{\mathcal{E}}^{\mathrm{in}}(u). \tag{4.7}$$

This relation holds for every group element u. An internal species direction on the matter side may change, with the corresponding unit-group charge change carried by the environment. If a reference carries coherence between charges, those components remain in the joint state and are assessed by total-charge sectors; the reference cannot be replaced by one definite charge value.

Rotation and angular momentum are treated by the same principle. If the actual matter, environment, and apparatus carry the appropriate spatial rotation representations, and the joint action commutes with the total rotation generator, the corresponding total angular-momentum component is preserved. Spatial motion and matter spin are carried by their respective representations, and their transfers enter the total together. Each matter mode uses its established spin representation [2], while motion orientation evolves under the generator of Chapter 2. A detector oriented along a given direction can exchange angular momentum, so its relevant degrees of freedom must be retained as well.

4.3 Balances in Events

Conservation constrains the complete evolution, while actual records determine which exchanged quantities can be read separately. A position record in Chapter 3 identifies the region where matter appears. To obtain the energy difference

in a particular exchange, the record must also resolve the corresponding energy outcomes.

In an energy-resolving recording process [1], a fresh auxiliary system is independent of the retained system, giving input $\rho \otimes \eta_{\mathcal{E}}$; memory that will participate again is included in the system. The auxiliary preparation is $\eta_{\mathcal{E}} = \sum_i \pi_i |i\rangle\langle i|$ in an energy eigenbasis, and its incoming and outgoing energies are ε_i and ε_o. The conditional operator that actually resolves both labels is $J_{oi} = \sqrt{\pi_i}\langle o|\mathbb{U}|i\rangle$. Joint conservation gives

$$\boxed{H_S J_{oi} - J_{oi} H_S = (\varepsilon_i - \varepsilon_o) J_{oi}.} \tag{4.8}$$

If a system matrix element from energy E_a to E_b is nonzero, then

$$E_b + \varepsilon_o = E_a + \varepsilon_i. \tag{4.9}$$

The system's energy increase equals the auxiliary's energy decrease component by component. Figure 4-1 places one such exchange alongside its record. For an input containing several energy components, each allowed transition still satisfies this relation; the actual record collects them according to the labels it resolves.

For outcome r with resolved energy difference q_r, the conditional action $\mathcal{I}_r$ defined in Chapter 3 can be represented by conditional operators $J_{r\ell}$, each satisfying $[H_S, J_{r\ell}] = q_r J_{r\ell}$. Let $F_r = \sum_\ell J_{r\ell}^\dagger J_{r\ell}$ be the corresponding effect operator, with $p_r = \mathrm{Tr}(F_r \rho) > 0$. Combining the commutation relation with its adjoint gives

$$\begin{aligned} [H_S, F_r] &= 0, \\ \sum_\ell J_{r\ell}^\dagger H_S J_{r\ell} &= H_S F_r + q_r F_r. \end{aligned} \tag{4.10}$$

Taking the energy trace of the output $\mathcal{I}_r(\rho) = \sum_\ell J_{r\ell} \rho J_{r\ell}^\dagger$ and dividing by the outcome probability yields

$$\boxed{\langle H_S \rangle_{\mathrm{out}|r} = \frac{\mathrm{Tr}(H_S F_r \rho)}{p_r} + q_r.} \tag{4.11}$$

The first term on the right is the incoming mean energy selected by the same outcome. Subtracting this selected incoming mean from the outgoing mean for the same outcome gives q_r. Directly subtracting the mean over all incoming matter from the selected outgoing mean would also include the effect of selecting different incoming components.

When a record groups several energy differences under one label, it generally determines no unique q_r. Its components must be retained, or the corresponding conditional means compared. A position label in Chapter 3 usually has this property. Conservation constrains all allowed outputs; resolving a definite exchange event by event depends on the record's resolution. Momentum and group-charge balances likewise use the outcomes actually resolved, without assigning complete unrecorded motion data to one position response.

Summing all outcomes along successive records adds the exchange balances. Intermediate contributions cancel, leaving the endpoints and net environmental and control changes. Selecting survival or a class of detection outcomes instead requires comparison along that conditional history; exits and unselected outcomes remain in the complete statistics.

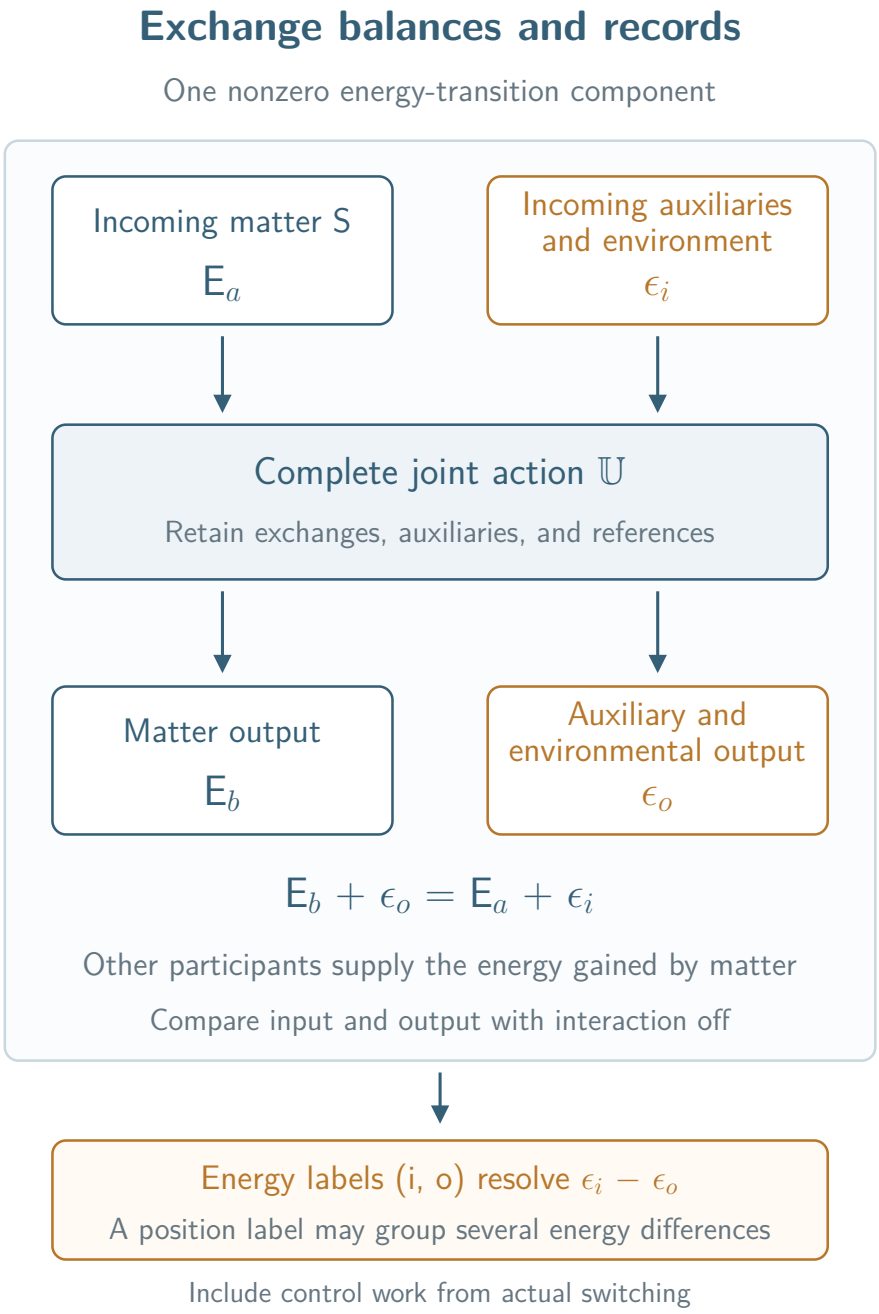


Figure 4-1. Balancing exchanged quantities and records. One nonzero energy-transition component is selected. The inputs and outputs of matter and all exchange auxiliary systems jointly satisfy total energy conservation. A record resolving the auxiliary input and output energies determines the energy gained by matter; a position-only record may contain several energy differences. The environment shown includes the apparatus, auxiliary systems, and references actually involved in this energy-conserving realization. Energy balances for externally scheduled control are calculated from the actual action as in Section 4.2.

Interference, tunneling, entanglement, and macroscopic trajectories further reveal how preparation, interaction, and recording alter motion.

§5 Phenomena of Motion

An individual electron leaves a discrete spot on a detector, whereas a stone appears to move along a continuous arc. From an electron's flight to a stone's fall, matter propagates while preserving its structure and leaves records through interactions with other matter. Propagation phases, distinctions preserved in interactions, and states following records jointly determine the distribution of spots and the conditions for continuous trajectories. Everyday motion is constituted by quantum motion.

5.1 Elementary Quantum Motion

An electron gun emits electrons one at a time into vacuum, and a distant detector responds successively. Each response leaves a position outcome; outcomes from many identical preparations form a position distribution. Changing the flight time moves or broadens this distribution. How the electron moves in flight and what the detector ultimately records must be connected within this complete process.

The electron preserves its internal structure during propagation [2], while its position amplitudes evolve under the dynamics of Chapter 2. Components at different positions jointly belong to that electron state. At the detector, the electron interacts with the apparatus, preserving distinctions between response regions and forming the records of Chapter 3. An absorbing screen may also terminate propagation of the original electron while leaving a spot.

Position amplitudes and their coherent evolution persist between departure and arrival. Without intermediate position records, a terminal spot does not supply a sequence of measured intermediate positions. With intermediate detection, the electron continues from the state left by that detection, changing the subsequent spots. Independent experiments with different flight times and successive tracking of the same electron therefore use different record histories.

The experimental statistics directly connect propagation in Chapter 2 with detection in Chapter 3. Let ρ_0 be the common preparation, $\mathcal{G}_t$ the actual evolution up to flight time t, and $\mathcal{I}_A$ the recording action for detector region A. Propagating first and then taking the share for that outcome gives

$$p(A;t) = \mathrm{Tr}\big[\mathcal{I}_A\big(\mathcal{G}_t(\rho_0)\big)\big] \,. \tag{5.1}$$

Changing the flight time compares terminal distributions at different propagation times. Experiments in which individual electron responses accumulate into an interference pattern connect local detection in a single run to statistics from repeated preparations [8].

Preparing an electron wave packet in a narrow region requires several wave-number components. The spatial phase of a single wave number extends around the entire position ring with constant modulus. Several components must add constructively in the preparation region and destructively elsewhere to form a localized distribution. Position and wave-number distributions are related by the Fourier expansion of the same amplitudes and cannot both be concentrated arbitrarily. The corresponding momentum uncertainty in the long-wavelength regime is already present in the preparation before the electron reaches the detector.

This restriction can be calculated directly in the finite position model of Chapter 2. Let $\psi_n = \chi_n(0)$ be the prepared position amplitudes and $\widehat{\psi}_k$ their discrete Fourier

transform. Let N_x and N_k count nonzero amplitude components in the position and wave-number representations of the normalized state, and let $L \geq 3$ be the number of positions. Every Fourier-matrix entry has modulus $L^{-1/2}$. Applying the Cauchy inequality to the position sum gives

$$\|\widehat{\psi}_k\|^2 \leq \frac{N_x}{L} \sum_n \|\psi_n\|^2 = \frac{N_x}{L}. \tag{5.2}$$

Summing over the wave numbers with nonzero amplitudes and using normalization yields

$$1 = \sum_k \|\widehat{\psi}_k\|^2 \leq \frac{N_k N_x}{L}, \qquad \boxed{N_x N_k \geq L.} \tag{5.3}$$

This is the support uncertainty relation for the finite Fourier transform [9]. With internal orientation, the same proof applies to vector norms. Restricting the electron to fewer position components requires more propagation components. Their phase relations determine the localized preparation before detection occurs.

Actual packets may retain small tails at many positions, so their concentration must also be measured by standard deviations. Using the uniform position calibration of Chapter 2 and the finite-difference momentum of Chapter 4, take

$$X = \ell_x \sum_{n=0}^{L-1} n|n\rangle\langle n|, \qquad \Pi_x = \frac{S_x^\dagger - S_x}{2i\ell_x}. \tag{5.4}$$

The position coordinate has a seam at $L-1 \leftrightarrow 0$. Write $W_\partial = |0\rangle\langle L-1|$. The position increment on an interior edge is ℓ_x, whereas crossing the seam gives $-(L-1)\ell_x$. The original shift therefore gives

$$\begin{aligned} [X, S_x] &= \ell_x S_x - L\ell_x W_\partial, \\ [X, S_x^\dagger] &= -\ell_x S_x^\dagger + L\ell_x W_\partial^\dagger. \end{aligned} \tag{5.5}$$

With $C_\ell = (S_x + S_x^\dagger)/2$ and $B_\partial = L(W_\partial + W_\partial^\dagger)/2$, substitution into the finite-difference momentum yields the exact finite-ring relation

$$\boxed{[X, \Pi_x] = i(C_\ell - B_\partial).} \tag{5.6}$$

For a normalized state at any specified instant, write $\langle A\rangle = \mathrm{Tr}(\rho A)$, set $\delta X = X - \langle X\rangle$ and $\delta\Pi_x = \Pi_x - \langle\Pi_x\rangle$, and define the symmetrized covariance $C_{X\Pi} = \frac{1}{2}\langle \delta X \delta\Pi_x + \delta\Pi_x \delta X\rangle$. Applying the Cauchy inequality to the positive semidefinite inner product $\langle A, B\rangle_\rho = \mathrm{Tr}(\rho A^\dagger B)$ gives

$$\begin{aligned} \sigma_X^2 \sigma_{\Pi_x}^2 &\geq |\langle \delta X \delta\Pi_x\rangle|^2, \\ \langle \delta X \delta\Pi_x\rangle &= C_{X\Pi} + \tfrac{i}{2}\langle C_\ell - B_\partial\rangle. \end{aligned} \tag{5.7}$$

The Schrödinger–Robertson relation with the covariance retained [10,11] thus takes the following form in the original finite model:

$$\boxed{\sigma_X^2 \sigma_{\Pi_x}^2 \geq C_{X\Pi}^2 + \tfrac{1}{4}\langle C_\ell - B_\partial\rangle^2.} \tag{5.8}$$

This also holds for mixed states and states with correlations between internal orientation and position; the position operators act as the identity on the other

factors. Let w_k and p_n be the wave-number and position populations of the state. Since $C_\ell(k) = \cos(k\ell_x)$, positivity gives

$$\begin{aligned} \varepsilon_k &= 1 - \langle C_\ell \rangle = \sum_k [1 - \cos(k\ell_x)] w_k, \\ \varepsilon_\partial &= L\sqrt{p_0 p_{L-1}}, \qquad |\langle B_\partial \rangle| \le \varepsilon_\partial. \end{aligned} \tag{5.9}$$

The first term measures the finite-wavelength correction, and the second controls the effect of the seam. Writing $a_+ = \max(a, 0)$, the preceding relations give a bound directly constraining the two widths:

$$\boxed{\sigma_X \sigma_{\Pi_x} \ge \tfrac{1}{2}(1 - \varepsilon_k - \varepsilon_\partial)_+.} \tag{5.10}$$

For a state whose wave-number support lies in $|k\ell_x| \le \delta$, with $0 < \delta < \pi/2$, one has $\varepsilon_k \le 1 - \cos\delta \le \delta^2/2$. A localized long-wavelength packet far from the seam therefore has a lower bound close to $1/2$. Both corrections must be controlled: a uniform zero-wave-number state has $\varepsilon_k = 0$, but its $\langle B_\partial \rangle = 1$ exactly cancels $\langle C_\ell \rangle = 1$, consistently with its zero momentum width.

Sample a smooth localized envelope on the lattice, decrease the spacing, and keep the seam far from its support. The central difference tends to $P = -i\partial_x$, while the position and momentum variances and their covariance converge, yielding

$$\boxed{\sigma_X^2 \sigma_P^2 \ge C_{XP}^2 + \tfrac{1}{4}, \qquad \sigma_X \sigma_P \ge \tfrac{1}{2}.} \tag{5.11}$$

Here $C_{XP} = \frac{1}{2}\langle \delta X \delta P + \delta P \delta X \rangle$ is the continuous-envelope covariance; restoring units gives the product bound $\hbar/2$. In the controlled continuum limit established above, this is precisely the position–momentum uncertainty principle. The position and momentum distributions of the same prepared state cannot both be arbitrarily concentrated: a smaller position standard deviation requires a larger lower bound on the momentum standard deviation. These standard deviations characterize the statistical distributions obtained by position and momentum readouts on identical preparations. The restriction also applies to pure states.

Position–momentum uncertainty is therefore already present in the prepared matter state before detection occurs. The original shift, finite-difference momentum, and positivity of the state jointly give the quantitative relation above. Appendix A.5 provides the sampling limit, the comparison with wave-number momentum, and control of the seam error during propagation. This relation constrains the prepared widths; subsequent phase evolution of the propagation components determines how those widths change in time.

After the electron leaves the source, its wave-number components evolve with their respective propagation phases. If their group velocities differ, components initially concentrated together gradually separate, allowing later detections to cover a wider region. The dispersion curvature of Chapter 2 determines this velocity difference: for a given momentum width, greater inertia gives a smaller velocity spread.

Consider a normalized surviving packet in the scalar return process of Chapter 2, on the positive-energy branch, with propagation remaining within the regime of the preceding continuous envelope and the quadratic dispersion approximation. Then $E(P) \simeq E_0 + P^2/(2M_{\text{in}})$. On the envelope, X and $P = -i\partial_x$ satisfy $[X, P] = i$, so uniform propagation gives $\dot{P} \simeq 0$ and $\dot{X} \simeq P/M_{\text{in}}$. Integration gives

$$X(t) \simeq X(0) + \frac{t}{M_{\text{in}}} P(0). \tag{5.12}$$

Subtracting expectations yields $\delta X(t) \simeq \delta X(0) + t\delta P(0)/M_{\text{in}}$. Squaring and taking the initial-state expectation, with C_{XP} denoting the initial symmetrized covariance, gives the position variance

$$\begin{aligned} \sigma_X^2(t) \simeq \sigma_X^2(0) + \frac{2t}{M_{\text{in}}} C_{XP} \\ + \frac{t^2}{M_{\text{in}}^2} \sigma_P^2(0). \end{aligned} \tag{5.13}$$

The three terms come from the original position variance, the initial position–momentum correlation, and velocity differences between momentum components. Without initial position–momentum correlation, the distribution of electron spots spreads with flight time. A correlated initial packet may first contract and then spread. A preparation spreading symmetrically in both directions can have no net displacement: its mean position stays fixed while the range of spots changes.

A cloud-chamber track consists of local interactions left along the particle's path. A charged particle passing through gas can ionize atoms or molecules; the ions promote condensation of supersaturated vapor, and tiny droplets amplify these distinctions into a visible track. A spherically symmetric outgoing preparation allows propagation in every direction, while successive positions within one track are correlated. This correlation must be obtained from the joint evolution of the particle and atoms.

Using the preceding low-speed continuous description, consider a neutral one-electron atom. Let $\mathbf{u}$ be the incident α particle's position relative to the atomic center of mass and $\mathbf{r}$ the electron's position relative to the nucleus. The cross term in the same charge energy [12] gives

$$V_{\alpha A}(\mathbf{u}, \mathbf{r}) = g \left[\frac{1}{|\mathbf{u} + \epsilon \mathbf{r}|} - \frac{1}{|\mathbf{u} - b\mathbf{r}|} \right]. \tag{5.14}$$

Here $g = 2\kappa$, with κ the electrostatic coefficient of the original charge coupling, $\epsilon = m/(M_N + m)$, and $b = M_N/(M_N + m)$; m and M_N are the electron and nuclear masses. The two terms describe the α particle's interactions with the nucleus and electron. Atomic center-of-mass recoil remains part of the same process.

The matrix element of this action between the initial and final atomic states,

$$W_{f0}(\mathbf{u}) = \langle f | V_{\alpha A}(\mathbf{u}, \mathbf{r}) | 0 \rangle. \tag{5.15}$$

determines the actual transition. For excitation from the spherically symmetric ground state to the three $2p$ states, the Fourier matrix elements obtained from the atomic wave functions give a summed transition strength that decreases with increasing momentum transfer. Excitation energy sets the minimum transfer; deflection from the forward direction increases it further and suppresses the transition.

For successive excitations of two atoms, the joint amplitude includes both orders of interaction. Each order joins the first transition, subsequent propagation under the original three-dimensional action, and the second transition. For the order through atom 1 first, the weak-collision, narrow-band, local far-field expansion over

the effective interaction region for heavy, slowly moving atomic centers gives the first transition's angular factor $\widetilde{W}_{f0}(\mathbf{q}) = \int d^3u\, W_{f0}(\mathbf{u}) e^{i\mathbf{q}\cdot\mathbf{u}}$, where

$$\begin{aligned} \mathbf{q} &= k_0\widehat{\mathbf{a}}_1 - k_1\widehat{\mathbf{a}_2 - \mathbf{a}_1}, \\ q^2 &= (k_0 - k_1)^2 + 4k_0k_1 \sin^2\frac{\theta}{2}. \end{aligned} \tag{5.16}$$

The source is at the origin, the atomic centers are at $\mathbf{a}_1, \mathbf{a}_2$, and θ is the angle between the two propagation segments. The actual energies before and after the transition determine k_0, k_1. The effective interaction range includes the transition charge's long-range tail. With small energy loss per encounter, k_1 is close to k_0, and larger excitation weights occur at smaller deflection angles. The second atom acts on this outgoing amplitude, favoring excitation positions along the same outward direction. Both interaction orders are added coherently, and the initial superposition of directions remains intact.

This gives the directional correlation of two local excitations, addressing the problem discussed by Mott [13]. In an actual cloud chamber, ionization and subsequent condensation further amplify local interactions into visible droplets; large or accumulated momentum exchanges bend the track. Successive interactions establish its direction without requiring the spherically symmetric preparation to select one ray in advance. The conditional histories of Chapter 3 describe these successive records, with all directions retained in the complete statistics. The actual transition matrix elements and the two-interaction expansion are derived in Appendix A.3.

A cloud chamber preserves distinctions along the path as a track. The double slit shows how the phase of two-path propagation affects terminal spots with and without path records.

5.2 Double-Slit Interference

An electron source emits electrons one at a time. After passing through two slits, they reach a screen. Each response leaves one spot, yet many spots accumulate into alternating bright and dark fringes [14]. Adding an interaction at the slits that preserves a complete path distinction removes the double-slit interference fringes. Both their formation and disappearance must be explained through the electron's passage through the apparatus.

From source to barrier, the electron propagates as in the preceding section. The barrier, path detectors, and screen are also made of matter. Local interaction with the barrier sends part of the amplitude into reflection, absorption, or other outputs; components transmitted through the openings continue to the screen, which preserves distinctions between response regions. Reflection, absorption, and no response all enter the complete statistics.

First take one coherent pure preparation, with both slits open and the two paths leaving a common state in all other unread degrees of freedom. For a mixed preparation, the corresponding probabilities are averaged over its preparation weights. Let $A_1(x), A_2(x)$ be the two path amplitudes arriving at screen position x. Together they give the response probability there:

$$\boxed{p_0(x) = |A_1(x) + A_2(x)|^2.} \tag{5.17}$$

Substituting $A_i = |A_i|e^{i\varphi_i}$ and setting $\Delta\varphi = \varphi_1 - \varphi_2$ gives

$$p_0(x) = |A_1|^2 + |A_2|^2 + 2|A_1||A_2| \cos \Delta\varphi(x). \tag{5.18}$$

The paths accumulate different phases during propagation. At some screen positions they add constructively, producing more spots; elsewhere they cancel, producing fewer. Calculating the actual fringe spacing requires the calibrated two-dimensional geometry and the relation between propagation phase and path length. The one-dimensional ring in Chapter 2 describes propagation along one direction and does not itself fix the experimental geometry. For approximately monochromatic incidence on a paraxial far-field screen, let d be the slit separation, L_s the screen distance, and k the wave number. The paraxial path difference is $\Delta r \simeq dx/L_s$; multiplication by k gives the phase difference. Adjacent bright fringes differ by another 2π, so

$$\Delta\varphi(x) \simeq \frac{kdx}{L_s}, \qquad \Delta x \simeq \frac{2\pi L_s}{kd}. \tag{5.19}$$

The phase relation of propagation thus determines fringe spacing. Closing a slit changes the apparatus, so the single-slit result must be calculated under the altered action. The cross term with both slits open comes from two paths in the same actual apparatus.

Adding path detection at the slits introduces another electron–detector interaction. For example, detector units near the openings may respond separately, preserving which-opening information in different pointers. With apparatus preparation $|D_0\rangle$, the two path components continue through while leaving respective records:

$$\begin{aligned} |1\rangle|D_0\rangle &\longmapsto |1\rangle|D_1\rangle, \\ |2\rangle|D_0\rangle &\longmapsto |2\rangle|D_2\rangle. \end{aligned} \tag{5.20}$$

Here D includes all unread internal, apparatus, and environmental degrees of freedom. Error-free path identification requires orthogonal record supports, or $\langle D_2|D_1\rangle = 0$ in this pure-state representation. The distinction meets the record criteria of Chapter 3 only if it remains after the complete action and is preserved while the electron continues toward the screen.

Both electron components can still reach the same screen position x, but they are correlated with records D_1 and D_2, respectively. Continue to denote the actual post-detection path amplitudes reaching the screen by A_1, A_2. The complete unread vector at that position is

$$|\Phi(x)\rangle = A_1(x)|D_1\rangle + A_2(x)|D_2\rangle. \tag{5.21}$$

The screen resolves only x. Summing over the other outputs is equivalent to taking the squared norm of this vector. Writing $\gamma(x) = \langle D_2|D_1\rangle$, expansion gives

$$\begin{aligned} p(x) &= \langle\Phi(x)|\Phi(x)\rangle \\ &= |A_1|^2 + |A_2|^2 \\ &\quad + 2\,\mathrm{Re}\big[A_1\overline{A_2}\gamma(x)\big]\,. \end{aligned} \tag{5.22}$$

The same cross term contains both the propagation phases of the paths and the overlap of the unread states. A complete path record gives $\gamma = 0$, hence

$$\boxed{p_{\mathrm{rec}}(x) = |A_1(x)|^2 + |A_2(x)|^2.} \tag{5.23}$$

The extra enhancement at some positions and extra suppression at others disappear, removing the double-slit fringes, while the diffraction distribution of each path may remain. The preserved distinction exists even if no one looks at the detector. Summing records does not restore a common apparatus state, so coherence in the complete joint state no longer appears as a cross term in this screen-position distribution.

With only partial path distinguishability, apparatus outputs overlap and the corresponding cross term remains on the screen. Over a fringe interval where the two magnitudes and $|\gamma|$ are approximately constant, constructive and destructive phases give maximum and minimum intensities. Their difference divided by their sum gives the visibility

$$\mathcal{V} = \frac{p_{\max} - p_{\min}}{p_{\max} + p_{\min}} = \frac{2|A_1||A_2||\gamma|}{|A_1|^2 + |A_2|^2}. \tag{5.24}$$

For equal path intensities, $\mathcal{V} = |\gamma|$. This relation expresses the constraint of path distinguishability on interference visibility [15]. Figure 5-1 compares the progression from indistinguishable paths to complete path records. Any phase change or momentum exchange produced by actual detection is also included in the path amplitudes.

If the apparatus can perform a complementary readout on all degrees of freedom preserving the distinction, the orthogonal pure records can instead be measured in the basis $|D_\pm\rangle = (|D_1\rangle \pm |D_2\rangle)/\sqrt{2}$. Projecting $|\Phi(x)\rangle$ onto these outcomes gives joint probabilities

$$p(x, \pm) = \tfrac{1}{2}|A_1(x) \pm A_2(x)|^2. \tag{5.25}$$

This complementary readout gives the conditional grouping of a quantum eraser [16]; coincidence detection of correlated photons provides an experimental realization [17]. The two groups have opposite interference terms, whose sum remains $|A_1|^2 + |A_2|^2$. Grouping can therefore reveal fringes without changing the unconditioned screen distribution. Record degrees of freedom that have been lost or cannot be controlled cannot be erased merely by a formal basis change. Coarse position readout or phase averaging can also reduce fringes and must be distinguished from actual path recording. In both double-slit cases, the screen leaves local spots one at a time; whether the two phases jointly affect spot probabilities depends on the path distinction left during propagation.

5.3 Tunneling

In a scanning tunneling microscope, a narrow vacuum gap separates the tip from a conducting sample. Applying a bias nevertheless produces a measurable current that is highly sensitive to gap width [18]. Incoming electrons have energy below the barrier height associated with the gap, yet a response appears on the far side. This transmission arises from complete electron propagation inside and outside the barrier.

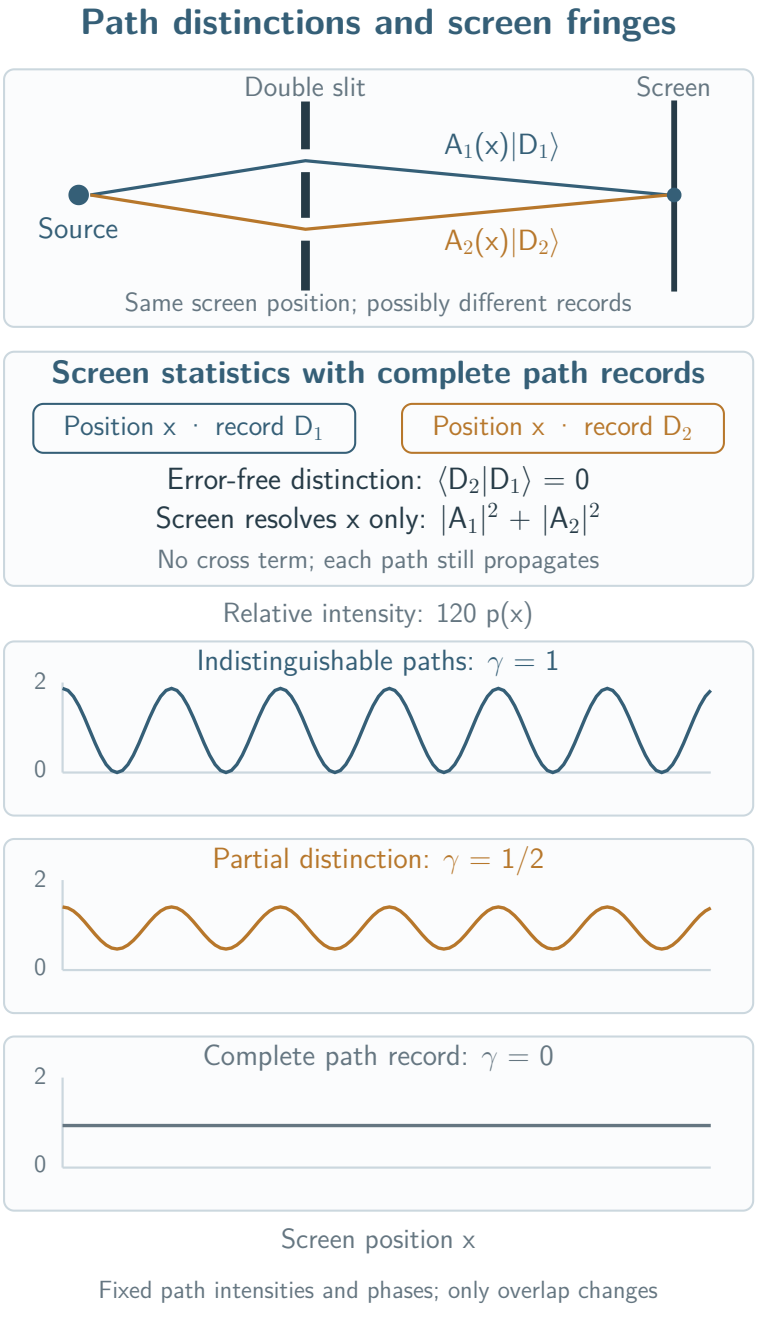


Figure 5-1. Path distinctions and screen fringes. Paths arriving at the same screen position may remain correlated with different records. Orthogonal complete records remove the path cross term when the screen sums over them. Below, path intensities and phases are held fixed while only overlap γ changes; the plotted distributions are $p_m = [1 + \gamma \cos(2\pi qm/L)]/L$, with $L = 120, q = 6$ and $\gamma = 1, 1/2, 0$, and each is normalized. The flat fringe-free curve comes from the equal-intensity example used here; an actual apparatus also retains the diffraction envelope of each path.

An electron wave packet reaches the barrier, interacts with matter or the background there, and separates into reflected and forward components. In the long-wavelength regime of the positive-energy branch of Chapter 2, an actual local energy change can be represented by a static potential $V(x)$. Background modulation of the maintenance frequency gives one class of local potential energy [2]; a specific apparatus uses its actual action. In the weak-potential, single-branch regime with the required spatial smoothness, the kinetic energy is $P^2/(2M_{\mathrm{in}})$. Differentiating the envelope's Fourier expansion term by term gives

$$-\partial_x^2 \sum_k a_k e^{ikx} = \sum_k k^2 a_k e^{ikx}. \tag{5.26}$$

The quadratic momentum term thus becomes $-\partial_x^2/(2M_{\mathrm{in}})$ in position space, while the local energy change multiplies the envelope at that position. Subtracting the common rest energy, the stationary electron envelope satisfies

$$\left[-\frac{1}{2M_{\mathrm{in}}}\frac{d^2}{dx^2} + V(x)\right]\psi(x) = \mathcal{E}\psi(x), \tag{5.27}$$

Here $\mathcal{E}$ is the incident energy relative to the rest energy. Outside the barrier, components oscillate with their propagation phase. On an approximately constant plateau of height $V_0 > \mathcal{E}$, rearrangement gives $\psi'' = \kappa^2\psi$, with $\kappa = \sqrt{2M_{\mathrm{in}}(V_0 - \mathcal{E})}$. The characteristic roots are $\pm\kappa$, so the complete solution on the plateau is

$$\psi(x) = B_+ e^{\kappa x} + B_- e^{-\kappa x}. \tag{5.28}$$

Both the incident wavelength and the decay length within the barrier must lie in this effective-motion regime; in particular, $\kappa\ell_x \ll 1$.

The share appearing on the far side requires matching the incident and reflected waves on the left, both barrier components, and the outgoing wave on the right. First calculate their probability current. For the effective equation above,

$$j = \frac{1}{M_{\mathrm{in}}}\,\mathrm{Im}(\psi^*\psi'). \tag{5.29}$$

Differentiating with respect to x, using that $\psi'^*\psi'$ is real, and substituting $\psi'' = 2M_{\mathrm{in}}(V - \mathcal{E})\psi$ for a real static potential gives

$$\partial_x j = \frac{1}{M_{\mathrm{in}}}\,\mathrm{Im}(\psi^*\psi'') = 0. \tag{5.30}$$

The current has the same value at every position. Substituting both exponential solutions on the plateau, the two squared-modulus terms are real and only the mixed terms contribute to the imaginary part:

$$j = -\frac{2\kappa}{M_{\mathrm{in}}}\,\mathrm{Im}(B_+^* B_-). \tag{5.31}$$

Either exponential alone has zero net current. Matching at both sides of a finite barrier determines the complex coefficients and their relative phase; the complete solution can carry nonzero current. The detector on the right reads this outgoing propagation output.

In a static elastic interaction, a transmitted electron retains its original energy and need not temporarily acquire energy above the barrier height. Without other

outputs, reflection and transmission together account for all incident current. Momentum associated with a change of direction is borne by the barrier and its supports. Absorption, inelastic exchange, or control work must be included in the corresponding balances of Chapter 4.

Now hold the incident preparation and both interfaces fixed while widening the plateau. Denote the linear matching across the actual transition regions on the two sides by C_L and C_R, determined by connecting solutions and derivatives of the same effective equation. Amplitudes on each side have their phases referenced to the respective interface. Across the plateau, the two exponentials acquire factors $e^{\kappa a}$ and $e^{-\kappa a}$. The total matching is

$$\begin{aligned} \mathsf{P}(a) &= \operatorname{diag}(e^{\kappa a}, e^{-\kappa a}), \\ \mathsf{M}(a) &= \mathsf{C}_R \mathsf{P}(a) \mathsf{C}_L. \end{aligned} \tag{5.32}$$

Take unit incident amplitude and reflected amplitude r on the left, and only outgoing transmitted amplitude t, with no incidence from the right. The endpoint conditions are

$$\begin{pmatrix} t \\ 0 \end{pmatrix} = \mathsf{M}(a) \begin{pmatrix} 1 \\ r \end{pmatrix}. \tag{5.33}$$

The second row gives $r = -\mathsf{M}_{21}/\mathsf{M}_{22}$; substitution into the first gives

$$t = \mathsf{M}_{11} + \mathsf{M}_{12} r = \frac{\det \mathsf{M}}{\mathsf{M}_{22}}. \tag{5.34}$$

With fixed interfaces, $d = \det \mathsf{M} = \det \mathsf{C}_R \det \mathsf{C}_L$ is independent of a, while $\mathsf{M}_{22} = \alpha e^{\kappa a} + \beta e^{-\kappa a}$, where α, β are determined by the interfaces. Thus

$$t(a) = \frac{d}{\alpha e^{\kappa a} + \beta e^{-\kappa a}}. \tag{5.35}$$

For $\alpha \neq 0$, $\kappa a \gg 1$, and $|\beta/\alpha| e^{-2\kappa a} \ll 1$, one has $t(a) \simeq (d/\alpha) e^{-\kappa a}$. If unit-amplitude incident and transmitted modes carry forward currents u_L, u_R, transmission is the current ratio $T = (u_R/u_L)|t|^2$, giving

$$\boxed{T(a) \simeq C e^{-2\kappa a}.} \tag{5.36}$$

Here $C = (u_R/u_L)|d/\alpha|^2$. Both the prefactor and exponent follow from the same two-sided matching. Increasing the plateau width by Δa multiplies the leading transmission by $e^{-2\kappa \Delta a}$. Figure 5-2 shows how a wider gap suppresses the far-side output. For a slowly varying barrier, away from turning points and where $|\kappa'(x)| \ll \kappa(x)^2$, the corresponding leading exponential is $\exp[-2 \int \kappa(x) dx]$; the end transitions remain determined by the same matching.

Many occupied incident states contribute to the tip–sample current. With bias, barrier properties, and other conditions fixed, increasing the gap reduces their transmission and decreases the current; bringing the tip closer increases transmission. The specific current still depends on incident occupations and the apparatus potential. Microscopic theory of the scanning tunneling microscope also relates the current to sample electron states near the tip [19]. Tip displacement changes the propagation connection between the two sides, changing the number of electrons collected on the far side. Decay inside the barrier thereby becomes visible through the current change.

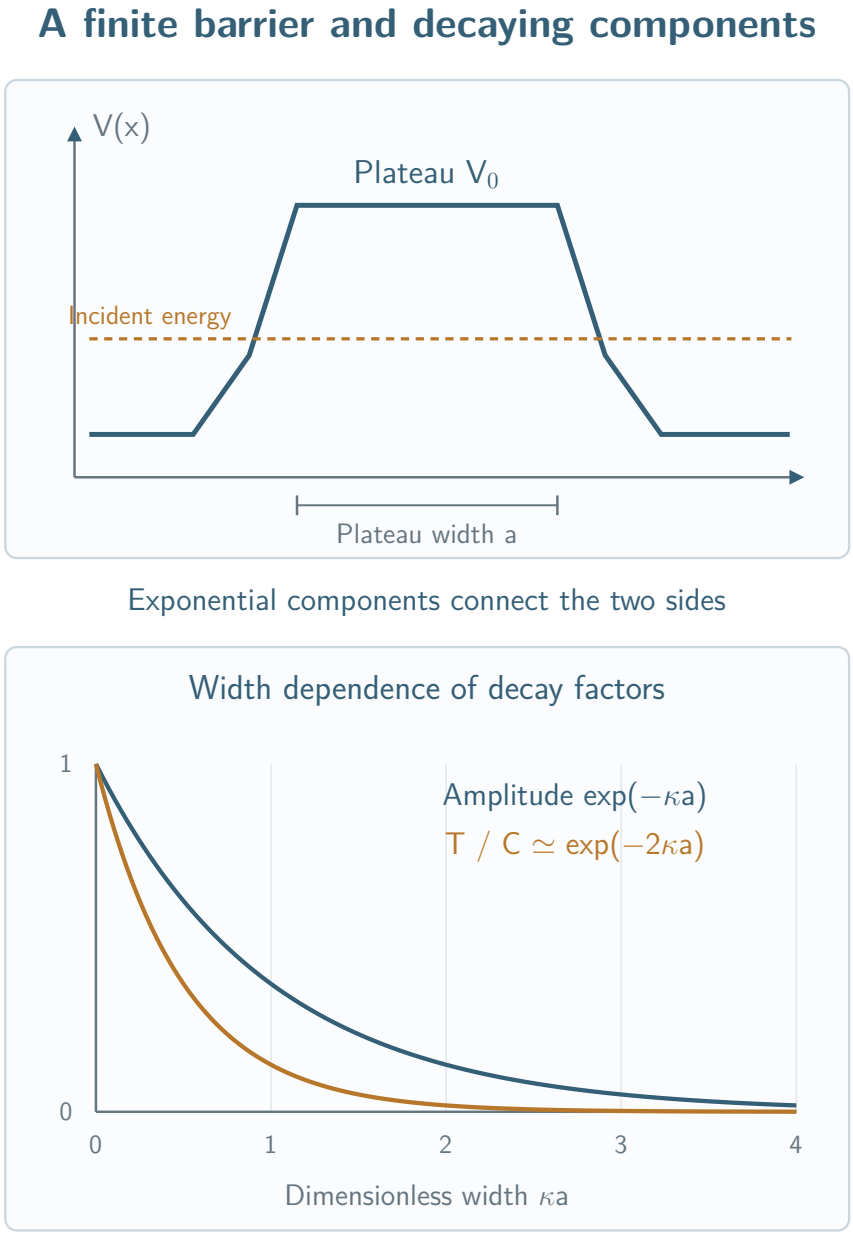


Figure 5-2. A finite barrier and decaying components. For incident energy below the plateau height, exponential components inside the barrier still participate in the matching between the sides. Below are the width dependences of the decaying amplitude and the leading exponential factor in thick-barrier transmission. A single decaying branch is not the complete transmitted solution; the actual current is determined by matching at both ends.

5.4 Entanglement

Two matter systems separate and propagate after a joint interaction, with outcomes recorded at each end. Neither end alone can read which operation the other has chosen. Correlations become visible when records from the two ends are paired by the same preparation.

A relay exchange produces such a pair through temporary storage, partial exchange, and inverse retrieval [2]. Take two persistent carriers A, B, each with accessible orthogonal modes $0, 1$, with each source and its auxiliaries in a fixed joint preparation that factorizes from the local logical input; pre-existing records remain spectators during this exchange. Let $\mathcal{J}_{AB}$ be their joint loading, and let mediator C start in $|u\rangle_C$. The actual complete action $\widehat{W}$ satisfies, on all two-level inputs,

$$\widehat{W}\mathcal{J}_{AB}(\psi \otimes u_C) = \mathcal{J}_{AB}(V_\theta \psi \otimes u_C), \tag{5.37}$$

where $\mathcal{J}_{AB}$ acts as the identity on the mediator, and V_θ preserves 00 and 11 while coherently exchanging 10 and 01. Input $|10\rangle$ gives

$$\boxed{|\Psi_\theta\rangle = \cos\theta|10\rangle + \sin\theta|01\rangle.} \tag{5.38}$$

Both 0 and 1 belong to the retained matter carrier; 0 does not denote exit. The exchange angle θ is determined by the actual strength and duration. The relay returns to a common preparation and retains no distinction between the two terms. When both coefficients are nonzero, this pure state cannot be factored into independent single-end states. The two matter systems then carry position packets toward different regions. Under local unitary evolution of the complete systems at the two ends, increasing distance does not remove entanglement, although correlations of internal readings remain affected by local motion and environmental action.

First take a coherent equal-amplitude pair, $\theta = \pi/4$. When both ends read modes $0, 1$, a 1 at one end accompanies a 0 at the other, with each allocation occurring half the time. A source randomly emitting either "10" or "01" would give the same anticorrelation. Identifying coherence between the two terms requires changing the readout.

Keep the pair preparation fixed and let both apparatuses read the coherent superpositions $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. When actual references and apparatus permit this readout, the coherent pair gives equal signs, whereas the randomly allocated mixture gives all four combinations.

The distinction follows directly by projection in the same readout. In the new basis, the coherent state is

$$\begin{aligned} |\Psi_+\rangle &= \frac{|10\rangle + |01\rangle}{\sqrt{2}} \\ &= \frac{|++\rangle - |--\rangle}{\sqrt{2}}. \end{aligned} \tag{5.39}$$

The two equal-sign amplitudes each have squared modulus $1/2$, and opposite-sign amplitudes vanish. The corresponding mixed preparation is

$$\rho_{\text{mix}} = \tfrac{1}{2}\big(|10\rangle\langle 10| + |01\rangle\langle 01|\big). \tag{5.40}$$

Each definite 0 or 1 gives either sign with probability one half in the ± readout. Multiplying single-end probabilities within each product state and summing with the mixture weights gives $p_{\mathrm{mix}}(s,t) = 1/4$. Thus the two preparations yield

Readout at both ends	Coherent equal-amplitude pair	Equal mixture of 10 and 01
Modes 0, 1	10 and 01, each with probability 1/2	10 and 01, each with probability 1/2
Modes +, −	++ and −−, each with probability 1/2	All four combinations, each with probability 1/4

After the apparatus change, either sign still occurs half the time at each end, but the pairing differs. The phase relation retained during joint propagation of the two terms is manifest in these joint records.

These readouts distinguish the coherent state from the particular mixture compared. Excluding general local classical models requires a fuller correlation test. Let each end choose between two measurement settings, randomly and independently of the source preparation. The four sets of paired statistics needed for a CHSH combination can then be compared [20,21]. Let σ_z read the original $0, 1$ modes and σ_x the ± superpositions, with actual settings

Setting	Two-outcome readout observable
First setting at A	$A_1 = \sigma_z$
Second setting at A	$A_2 = \sigma_x$
First setting at B	$B_1 = (-\sigma_z + \sigma_x)/\sqrt{2}$
Second setting at B	$B_2 = (-\sigma_z - \sigma_x)/\sqrt{2}$

In $|\Psi_+\rangle$, the original modes always give opposite signs, the superposition modes always give equal signs, and cross-basis products have zero mean. Applying these operators gives

$$\begin{aligned} \langle \sigma_z \otimes \sigma_z \rangle = -1, \quad \langle \sigma_x \otimes \sigma_x \rangle = 1, \\ \langle \sigma_z \otimes \sigma_x \rangle = \langle \sigma_x \otimes \sigma_z \rangle = 0. \end{aligned} \tag{5.41}$$

Write $E_{ab} = \langle A_a \otimes B_b \rangle$. Substituting the table's linear combinations into these results gives the four means

$$(E_{11}, E_{12}, E_{21}, E_{22}) = \frac{(1, 1, 1, -1)}{\sqrt{2}}. \tag{5.42}$$

This actual matter pair therefore yields

$$\boxed{S = E_{11} + E_{12} + E_{21} - E_{22} = 2\sqrt{2},} \tag{5.43}$$

For comparison, if each pair carries jointly assigned classical values $a_1, a_2, b_1, b_2 \in \{+1, -1\}$, the corresponding expression in one run is

$$C = a_1(b_1 + b_2) + a_2(b_1 - b_2). \tag{5.44}$$

If b_1, b_2 have equal signs, the second bracket vanishes; if they have opposite signs, the first vanishes. The other bracket has magnitude two, so $|C| = 2$ in each run. Averaging over any joint distribution gives

$$|S| = |\langle C \rangle| \leq \langle |C| \rangle = 2. \tag{5.45}$$

The CHSH combination of these four quantum correlations exceeds this bound. In terms of equal- and opposite-sign proportions, each reading product is ± 1, so $P_{\text{same}} = (1 + E_{ab})/2$. Equal signs in the first three setting pairs and opposite signs in the fourth pair each occur with proportion $(2 + \sqrt{2})/4$, about 85.4%. A joint classical assignment can satisfy at most three of the four requirements, giving at most 75% when the four setting pairs are sampled equally. This comparison uses one matter preparation and specified readout settings. Actual phase changes, detection losses, and no-response outcomes must be included in the experimental scheme; favorable detected samples cannot be selected according to setting. Experiments with spatially separated electron spins have tested such Bell correlations while simultaneously addressing locality and detection loopholes [22].

Viewed separately, the complete outcome sequence at A cannot identify the measurement setting chosen at B. The latter may group its records by outcome, producing subsamples correlated with A. To identify these groups, however, A must receive the records from B and pair them by the same preparation. Summing over all B outcomes leaves the state at A unchanged.

This follows directly from the complete instruments of Chapter 3. Denote the joint state after separated propagation by ρ_{AB}, and the readout actions by $\mathcal{I}_r^a$ and $\mathcal{J}_s^b$. The actual paired probability is

$$p(r, s \mid a, b) = \mathrm{Tr}\left[(\mathcal{I}_r^a \otimes \mathcal{J}_s^b)(\rho_{AB})\right]. \tag{5.46}$$

Combine all outcomes at B and denote its action operators by K_ν. Completeness requires $\sum_\nu K_\nu^\dagger K_\nu = I$. For any readout effect F_r at A, cyclicity of the trace gives

$$\begin{aligned} p'_A(r) &= \sum_\nu \mathrm{Tr}\left[(F_r \otimes K_\nu^\dagger K_\nu)\rho_{AB}\right] \\ &= \mathrm{Tr}[(F_r \otimes I)\rho_{AB}] \\ &= p_A(r). \end{aligned} \tag{5.47}$$

Since the equality holds for every effect at A, its nonselective reduced state is preserved. Retaining only one outcome s at B generally gives $\sum_{\nu \in s} K_\nu^\dagger K_\nu \neq I$, so conditional probabilities may change. Identifying that subsample still requires communication of the outcome label from B.

Entanglement correlations appear in paired records, while the complete statistics at either end remain unchanged.

5.5 Macroscopic Motion

A stone released from the hand falls along an arc; everyday observation shows one object moving along a continuous trajectory. The stone has many internal constituents, while surrounding light and air continuously interact with it. Collective motion and a continuous trajectory must be explained through preservation of internal structure, position propagation, and successive records together.

A composite object that preserves its internal structure over the interval considered can change its overall position while maintaining the relative relations of its constituents. Take a collective propagation branch satisfying the complete-carrier preservation relation [2], and consider collective propagation in its low-momentum regime. Internal actions and binding jointly determine the actual spectrum $E_{\mathrm{body}}(P)$, with inertia read from its curvature at the origin. The collective spectrum of a specific body must still be obtained from its constituents and interactions.

For an even, smooth positive-energy branch with a nondegenerate local minimum at the origin, expansion gives positive collective inertia and kinetic energy:

$$\begin{aligned} \frac{1}{M_{\mathrm{body}}} &= E''_{\mathrm{body}}(0), \\ E_{\mathrm{body}}(P) &= E_0 + \frac{P^2}{2M_{\mathrm{body}}} + O(P^4). \end{aligned} \tag{5.48}$$

Collective motion includes the common displacement of the constituents and the response of the field maintaining their binding. Consider a spherically symmetric, weakly electrostatically bound two-constituent state. Write the sum of the separated constituent masses as $M = m_1 + m_2$ and the binding energy as $I > 0$. With the common matter–field speed calibration $c_m = c_f = c_0$, retain the leading binding correction. The constituent motion spectrum and transverse-current interaction [12] respectively give

$$\begin{aligned} \delta E_{\mathrm{kin}} &= -\frac{5I}{6M^2c_0^2}P^2, \\ \delta E_{\mathrm{current}} &= \frac{4I}{3M^2c_0^2}P^2. \end{aligned} \tag{5.49}$$

Adding both terms to the original center-of-mass kinetic energy gives

$$E_C(P) \simeq Mc_0^2 - I + \frac{P^2}{2M}\left(1 + \frac{I}{Mc_0^2}\right). \tag{5.50}$$

Thus, to this order,

$$\boxed{\begin{aligned} E_C(0) &\simeq Mc_0^2 - I, \\ M_{\mathrm{in}} &\simeq M - \frac{I}{c_0^2}. \end{aligned}} \tag{5.51}$$

Binding energy enters both rest energy and spectral curvature. The mass deficit here is calculated jointly from constituent motion and the binding field; the transverse current is an essential contribution. The derivation is given in Appendix A.1.

For a given momentum width, larger inertia brings the velocities of propagation components closer together and can slow spreading between records. This is the same mechanism as electron spreading in Section 5.1. Whether the stone remains within a narrow position range over the interval considered also depends on its preparation and its continuing environmental interactions.

The cloud chamber already shows that a single particle can leave a nearly straight track through successive ionizations. A stone's macroscopic trajectory likewise requires the environment to preserve position distinctions. The spreading and recoil

of the entire composite object between records must also be considered; together they determine whether neighboring records can form a smooth history of motion.

Light scattered by the stone produces outputs that depend on its position region. If distinctions between different positions remain after scattering and are preserved, they become position records under the criteria of Chapter 3. Summing environmental outcomes can reduce interference between positions [23]; reading one outcome gives a compatible object state. Its position range narrows only if the actual action has sufficient position resolution.

A continuous trajectory comes from joining the state after each record to the next propagation stage. A position record narrows the conditional distribution, subsequent free propagation spreads it, and the same detection transfers recoil to the object and its reference.

Take a finite position instrument that preserves the selected motion branch and reads position relative to the reference. In the low-speed Gaussian envelope before wraparound, let s be the Gaussian pointer's noise variance and μ the reduced mass of the object and reference. Write the relative position variance, symmetrized position–momentum covariance, and momentum variance after one record as a, b, d. After free propagation for τ, with $h = \tau/\mu$,

$$\begin{aligned} a_- &= a + 2hb + h^2 d, \\ b_- &= b + hd, \qquad d_- = d. \end{aligned} \tag{5.52}$$

The next actual record gives

$$\boxed{\begin{aligned} a' &= \frac{s\, a_-}{a_- + s}, \\ b' &= \frac{s\, b_-}{a_- + s}, \\ d' &= d_- - \frac{b_-^2}{a_- + s} + \frac{1}{4s}. \end{aligned}} \tag{5.53}$$

The first relation shows narrowing after a record; the term $1/(4s)$ in the last is the back-action of the same detection. Improving position resolution increases momentum disturbance, with both determined by one instrument. Here τ counts only free propagation; the complete recording cycle also includes actual control and readout durations.

Within the pure Gaussian family, this recursion has a stable conditional width, derived together with the actual instrument and recoil energy in Appendix A.2. When the maximum width between records and accumulated recoil remain within the required resolution, neighboring records continuously identify the same object's motion relative to the reference. Centers still differ between recorded histories, and the distribution summed over all outcomes can continue to spread. The trajectory in Figure 5-3 corresponds to the concentrated state continued along actual records.

The environment also exchanges momentum. A large collision can deflect the stone, and small random recoils can accumulate. Increasing the recording frequency may also change propagation. A concentrated conditional history is suitably described by a smooth trajectory only when displacements caused by these actions are small on the resolution scale used. Mass, environmental records, and actual recoil

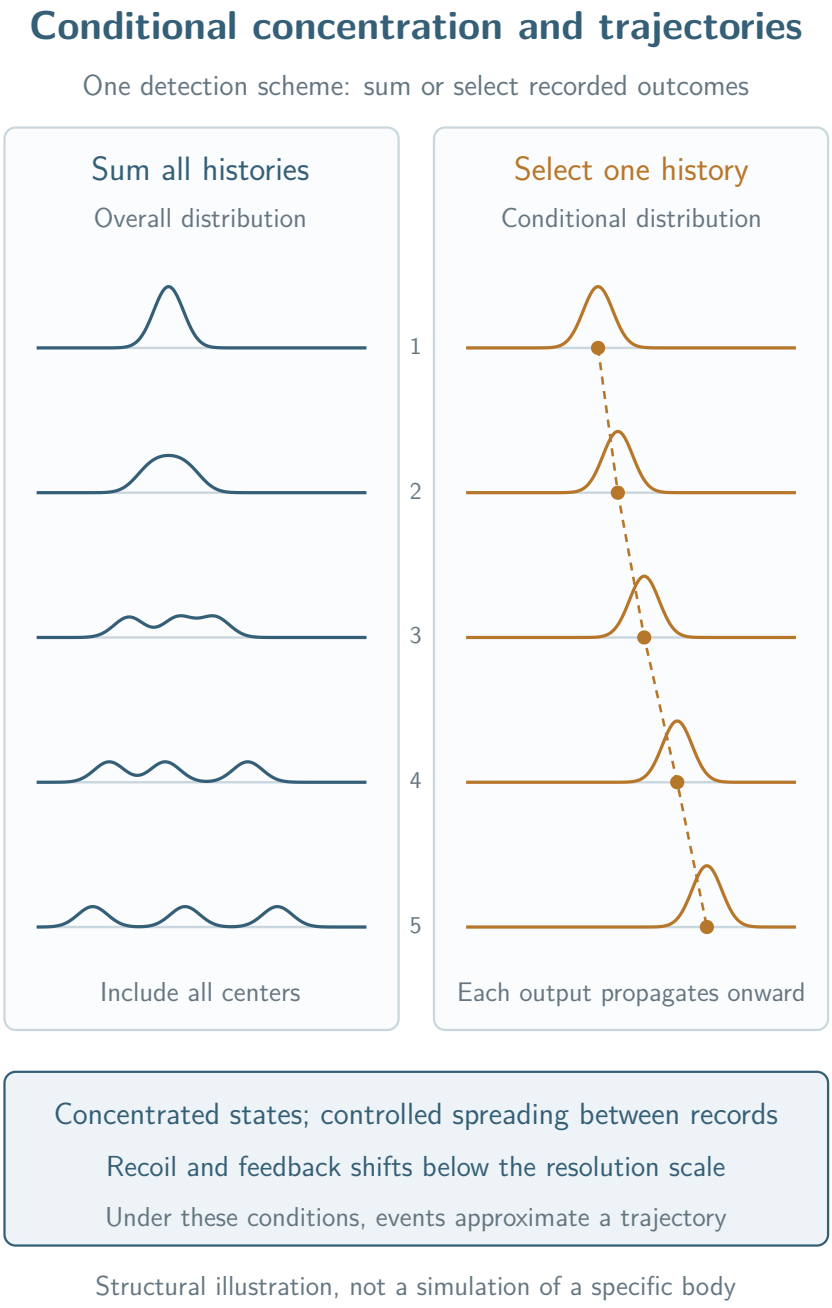


Figure 5-3. Conditional concentration and a macroscopic trajectory. Within the same detection scheme, the left panel sums all histories, while the right continues along one obtained record sequence. Successive outputs on the right feed into one another, and the concentrated region moves with the body. The thin line approximates this motion history when spreading and recoil are controlled. This is a structural illustration, not a numerical simulation of a specific body.

jointly determine this regime; localization and back-action in continuous measurement jointly constrain the emergence of classical trajectories [24].

After release, the stone retains horizontal momentum if drag and random recoil are negligible, while gravity changes its vertical momentum and gradually deflects its motion downward. This center motion can be calculated from the same low-speed envelope. Between records, take $H = E_0 + P^2/(2M_{\mathrm{body}}) + V(X)$. Within the effective continuous approximation, write $X_c = \langle X \rangle$ and $P_c = \langle P \rangle$. The Ehrenfest relations [25] give

$$\begin{aligned} \dot{X}_c &= i\langle [H, X] \rangle = \frac{P_c}{M_{\mathrm{body}}}, \\ \dot{P}_c &= i\langle [H, P] \rangle = -\langle V'(X) \rangle. \end{aligned} \tag{5.54}$$

For a narrow conditional state, expand $V'(X)$ about X_c. The linear term has zero expectation, and the quadratic remainder is controlled by the width:

$$\langle V'(X) \rangle = V'(X_c) + O\big(\sigma_X^2 \sup |V'''|\big)\,. \tag{5.55}$$

If the potential varies slowly over the packet scale, the latter term is negligible, giving $M_{\mathrm{body}}\ddot{X}_c \simeq -V'(X_c)$. In terms of the local energy $E(P, X)$ valid in this regime, the two center equations are

$$\boxed{\dot{X}_c \simeq \frac{\partial E}{\partial P}, \qquad \dot{P}_c \simeq -\frac{\partial E}{\partial X}.} \tag{5.56}$$

The right-hand sides are evaluated at (P_c, X_c). The first equation gives the motion of the stone's center; the second gives the change of its momentum under the local action.

The same bound object can be connected directly to its weak-background response. Let Φ denote the common relative change in maintenance frequencies, with displacement coefficients and electrostatic strength held fixed [2], with the background slowly varying over the bound object. Internal kinetic energy changes with maintenance, and the binding calculation gives

$$E_C(0, \Phi) \simeq (1 + \Phi)(Mc_0^2 - I). \tag{5.57}$$

The background potential and inertia in this example therefore use the same rest energy after binding:

$$\begin{aligned} V(X) &\simeq E_C(0)\Phi(X), \\ M_{\mathrm{in}}c_0^2 &\simeq E_C(0). \end{aligned} \tag{5.58}$$

Substitution into the center equations above gives

$$\ddot{X}_c \simeq -c_0^2 \Phi'(X_c). \tag{5.59}$$

The binding contribution is included in both coefficients and cancels from the acceleration to this order. This realizes the background response of the collective branch in an example with the constituents and binding field participating together.

For approximately uniform near-surface gravity, let $g_{\mathrm{grav}} > 0$ denote the magnitude of downward acceleration and take the z axis upward. Integrating the horizontal and vertical equations gives

$$
\begin{aligned}
x_c(t) &= x_0 + v_{x0}t, \\
z_c(t) &= z_0 + v_{z0}t - \tfrac{1}{2}g_{\mathrm{grav}}t^2.
\end{aligned}
\tag{5.60}
$$

Eliminating t gives the falling arc produced jointly by horizontal motion and vertical acceleration. Successive environmental records make the history continuously identifiable. Drag or collisions alter the arc through actual momentum exchanges. The stone's internal structure remains maintained, its collective position continues to propagate, and the environment leaves new position distinctions. These processes jointly constitute the continuous motion of everyday observation.

Discrete electron spots, cloud-chamber tracks, double-slit fringes and their disappearance, responses beyond a barrier, correlations between distant records, and a stone's arc show the same matter process under different apparatus and recording conditions. Quantum propagation determines states between interactions; successive records appear as macroscopic trajectories in an appropriate regime of concentration.

§6 Motion and Light

As a stone falls, incident light scatters into its surroundings. Some enters a camera and leaves responses on its photosensitive surface. The stone changes position, and successively arriving light changes the image. This actual process connects the matter motion of Chapter 5 to distant observation. Light propagation, exchange with matter, and records preserved at the receiver jointly provide the conditions under which we see motion.

6.1 Light Propagation

After leaving an emitter, light can cross the vacuum between two apparatuses and reach the other end. Emitter and receiver each preserve their matter structure, while light carries energy and state distinctions between them. The internal photon direction is determined by a zero mode of the neutral response [2]. Let B_Z be the actual response column and $|\gamma\rangle$ the selected direction. The complete input–output action satisfies

$$B_Z|\gamma\rangle = 0, \qquad \mathbb{H}_B \begin{pmatrix} \gamma \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & B_Z^\dagger \\ B_Z & 0 \end{pmatrix} \begin{pmatrix} \gamma \\ 0 \end{pmatrix} = 0. \tag{6.1}$$

This internal direction has no rest half-gap under the condensation mass action used. It is uncharged under the unbroken Abelian channel, while still exchanging energy and momentum with charged endpoints.

The spatial action preserving this internal direction is supplied by connecting the three-dimensional local source to transverse propagation [12]. In a uniform deployment, let ℓ be the logical spacing and $\bar{\nu}$ the exchange strength read per actual elapsed time. The positive-frequency transverse branch is

$$\omega_\gamma(\mathbf{k}) = \bar{\nu}\sqrt{\sum_{i=1}^{3} \sin^2(k_i \ell)}. \tag{6.2}$$

Its long-wavelength expansion gives

$$\omega_\gamma(\mathbf{k}) \simeq c_\gamma |\mathbf{k}|, \qquad c_\gamma = \bar{\nu}\ell. \tag{6.3}$$

One position action determines long-wavelength propagation in all three directions. For wave number along the x axis, the transverse amplitudes are (E_y, E_z, B_y, B_z). The original curl exchange separates, under the following wave-number-independent change of basis, into two orientations, each containing two polarizations:

$$\varphi_\pm = \frac{1}{\sqrt{2}} \begin{pmatrix} E_y \pm B_z \\ E_z \mp B_y \end{pmatrix}, \qquad i\partial_t \varphi_\pm = \pm \bar{\nu} \sin(k\ell) \varphi_\pm. \tag{6.4}$$

This relation follows directly from the transverse exchange matrix in Appendix A.4. The positive-energy branch is selected according to the sign of $\sin(k\ell)$. Ordering φ_+ before φ_-, the actual propagation in this basis is $\bar{\nu}\sin(k\ell) Z_{\mathrm{mov}} \otimes I_{\mathrm{pol}}$.

Using the spacing $\ell_x = \ell$ of Chapter 2 and matching the actual propagation strength through $\bar{\nu} = f_{\mathrm{act}} g$ gives $c_\gamma = \bar{\nu}\ell = c_0$. Appendix A.4 supplies the spatial

correspondence and coefficient relation between the field return and this propagation. In the active-time representation of Section 2.5, the same transverse action becomes

$$h_\gamma(k) = gZ_{\mathrm{mov}} \sin(k\ell_x) \otimes I_{\mathrm{pol}}. \tag{6.5}$$

Amplitude exchange between neighbors continues. Taking $Z_{\mathrm{mov}} u_\pm = \pm u_\pm$, direct application gives active eigenvalues $\epsilon_{\gamma,\pm} = \pm g \sin(k\ell_x)$. Selecting the positive-energy branch and multiplying by f_{act} for the complete-cycle timing of Chapter 2 gives

$$\boxed{E_\gamma(k) = f_{\mathrm{act}} g \, |\sin(k\ell_x)|.} \tag{6.6}$$

Motion orientation contributes to the position current, propagation direction is determined jointly by the selected wave number and branch, and polarization describes the transverse amplitudes within each orientation. The half-gap at $k = 0$ is zero, while the propagation speed is determined by how the phase varies with wave number.

Take a narrow packet whose wave-number support lies near, but entirely on one side of, $k = 0$, before spatial wraparound. Differentiating the spectrum for $0 < |k\ell_x| < \pi/2$ gives the group velocity and spectral curvature

$$\begin{aligned} v_g(k) &= \mathrm{sgn}(k) c_0 \cos(k\ell_x), \\ E''_\gamma(k) &= -c_0 \ell_x |\sin(k\ell_x)|. \end{aligned} \tag{6.7}$$

Their roles in pulse motion follow directly from the position amplitude. Let the narrow packet be centered at k_0, write $q = k - k_0$, and take a fixed orientation-and-polarization vector u_0 for that direction. Propagation is

$$\psi_\gamma(x,t) = u_0 \sum_q a_q e^{i(k_0+q)x - iE_\gamma(k_0+q)t}. \tag{6.8}$$

Expand the energy in the phase as

$$\begin{aligned} E_\gamma(k_0+q) = {} & E_\gamma(k_0) + v_g(k_0) q \\ & + \tfrac{1}{2} E''_\gamma(k_0) q^2 + O(q^3). \end{aligned} \tag{6.9}$$

Factoring out the common carrier phase and setting $\xi = x - v_g(k_0)t$ leaves the envelope

$$F_t(\xi) = \sum_q a_q e^{iq\xi - \frac{i}{2} E''_\gamma(k_0) q^2 t + \cdots}. \tag{6.10}$$

The linear phase translates the original envelope to $x - v_g t$, while the quadratic phase varies with q and further deforms it. In the long-wavelength regime, the sine spectrum expands as

$$E_\gamma(k) = c_0 |k| \left[1 - \frac{(k\ell_x)^2}{6} + O((k\ell_x)^4) \right], \tag{6.11}$$

Thus, to leading order,

$$E_\gamma(k) \simeq c_0 |k|, \qquad |v_g| \simeq c_0. \tag{6.12}$$

The emitted pulse therefore moves approximately as a whole through this uniform region. Changing the central frequency within the long-wavelength regime

leaves the leading-order group velocity unchanged. The finite lattice spectrum retains higher-order variation, so velocity differences between components must be included over sufficiently long propagation or broader wave-number ranges. Nearly linear longitudinal dispersion also does not remove transverse diffraction: a beam of finite width contains components in different spatial directions and can spread during propagation.

Photon polarization uses a transverse vector representation [2]. For the three-dimensional lattice action above, set $d_i(\mathbf{k}) = \sin(k_i \ell)$. When $\mathbf{d}(\mathbf{k}) \neq 0$, transverse amplitudes obey

$$\mathcal{T}_{\mathbf{k}} = \{\varepsilon \in \mathbb{C}^3 : \mathbf{d}(\mathbf{k}) \cdot \varepsilon = 0\}, \qquad \dim_{\mathbb{C}} \mathcal{T}_{\mathbf{k}} = 2. \tag{6.13}$$

At long wavelengths, $\mathbf{d} \simeq \ell\mathbf{k}$ recovers $\mathbf{k} \cdot \varepsilon = 0$. The two independent components perpendicular to propagation carry the two polarizations; their coherent combinations give linear, circular, and general elliptical polarization. During uniform polarization-preserving propagation, light reaches the receiver with this transverse state. The receiver may respond differently to different polarizations, and the share actually received is determined by that interaction.

The same light pulse thus has a propagation direction, frequency distribution, and polarization state. These arrive at another matter system and enter its next exchange. The spatial distribution specifies possible response positions; the receiver's action converts the receivable part into a change of matter state.

6.2 Light–Matter Interaction

Light acquires energy from an emitter, may scatter from an object in transit, and may be absorbed by a receiver. Emission, scattering, and absorption are joint exchanges between light and matter. Reflected, transmitted, and other outputs continue propagating with the amplitudes and phases formed by the interaction.

The interaction between emitter and light first produces a new field amplitude. In the closed-loop excitation mechanism [12], a downward matter transition creates a loop near the source, and local contact connects it to spatial propagation. Creation, outward propagation, and return to the source all participate in the evolution.

Let $\Delta > 0$ be the energy difference of the downward transition, A the actual source-vertex matrix element, and λ the source strength. Write $|r\rangle$ for the loop state near the source and let K include its complete connection to the propagation space. The energy difference and matrix element are determined by the actual spectrum and contact of the finite closed-loop source in Appendix A.4. At first order in the weak source, the outgoing component after source activity for T is

$$|\chi_T^{(1)}\rangle = -i\lambda A P_\gamma \int_0^T e^{-iK(T-t)} |r\rangle e^{-i\Delta t}\, dt. \tag{6.14}$$

Here P_γ selects the component in the propagation space. Each term in the integral corresponds to a creation time: matter makes the downward transition then, and the field amplitude propagates until T. Different creation times add coherently. For joint propagation frequency ε, the time integral contains

$$T\, e^{-i(\varepsilon+\Delta)T/2} \operatorname{sinc}\left[\frac{(\varepsilon-\Delta)T}{2}\right]. \tag{6.15}$$

Here $\operatorname{sinc} x = \sin x/x$, defined by continuity at zero. Components close to the source energy difference accumulate amplitude, while detuned components undergo phase cancellation. The main peak narrows as the interaction duration increases; actual output also depends on the source's coupling to propagation modes. In the continuous-spectrum window with boundary returns controlled, the original coupling has nonzero propagation weight near Δ, so the frequency selection acts on modes that can actually be excited. The spectral location of the source energy difference and the spatial correspondence are derived in Appendix A.4.

Emitter recoil, field energy, and energy exchanged when the interaction is switched on or off are still calculated within one process as in Chapter 4. Let H_{mat} be the energy of the emitter and participating supports, and H_γ^{ex} the light energy in the actual exchange. For a complete energy-resolved exchange $\mathbb{U}$ with no external work and with interaction off at the input and output, joint conservation gives the following relations.

Let i, o be definite incoming and outgoing light-energy labels with energies $\varepsilon_i, \varepsilon_o$, including a vacuum label of zero energy. The corresponding matter action is $J_{oi} = \langle o|\mathbb{U}|i\rangle$. Taking the matrix element of the joint commutation relation between the light states $|i\rangle$ and $|o\rangle$ gives

$$\begin{aligned} [\mathbb{U}, H_{\mathrm{mat}} + H_\gamma^{\mathrm{ex}}] &= 0, \\ [H_{\mathrm{mat}}, J_{oi}] &= (\varepsilon_i - \varepsilon_o) J_{oi}. \end{aligned} \tag{6.16}$$

Taking a further matrix element between matter energy eigenstates a, b yields

$$(E_b + \varepsilon_o - E_a - \varepsilon_i)\langle b|J_{oi}|a\rangle = 0. \tag{6.17}$$

An actual nonzero transition requires the bracket to vanish. For emission, incoming light is vacuum and outgoing light has energy E_γ^{ex}, so

$$E_{\mathrm{mat}}^{\mathrm{in}} = E_{\mathrm{mat}}^{\mathrm{out}} + E_\gamma^{\mathrm{ex}}. \tag{6.18}$$

When this exchange energy and propagation frequency have been compatibly calibrated, $E_\gamma^{\mathrm{ex}} = \omega$ in units $\hbar = 1$. With recoil, total matter energy also includes motion, so the internal-level difference must supply recoil energy as well. For absorption, outgoing light is vacuum, and the same nonzero-transition condition gives the matter energy increase. Once a receiver state that can be retained has formed, propagation of the original incident light may terminate.

In scattering, both incoming and outgoing light usually exist, but direction, frequency, or polarization can change. In Compton scattering, the light-frequency shift is linked to matter recoil, exhibiting this energy–momentum exchange [26]. For a realization with the actual additive momentum conservation of Chapter 4, if incoming and outgoing channels also resolve photon momenta p_i, p_o, the same matrix-element calculation gives

$$[P_{\mathrm{mat}}, J_{oi}] = (p_i - p_o) J_{oi}. \tag{6.19}$$

Thus a nonzero transition gives matter $\Delta P_{\mathrm{mat}} = p_i - p_o$. If mirror velocity changes are negligible and reflection is approximately elastic, the normal photon momentum changes from p_n to $-p_n$, transferring impulse $2p_n$ to the mirror and supports. The displacement of a macroscopic mirror may be tiny, yet the momentum transfer remains part of the complete process. Successive momentum exchanges by

many photons produce radiation pressure, whose mechanical effects can be measured directly [27]. Light can therefore both carry away information about motion and alter subsequent motion of the illuminated object.

Coherent reflection can preserve phase relations between incident components. Whether emission, scattering, or absorption forms a record is determined from the complete output using Section 3.1.

Absorptive photodetection converts photon absorption into a countable response through interaction with the receiver [28]. On a camera sensor, absorption can change local charge or another readable state; subsequent action preserves the response in the corresponding pixel. Each reported camera outcome comes from this receiving process. For repeated identical preparations with at most one selected response per cycle, let p_{det} be the response probability and τ_{tot} the complete cycle duration. The expected count in N cycles is Np_{det}; division by total elapsed duration gives

$$\overline{R}_{\mathrm{det}} = \frac{Np_{\mathrm{det}}}{N\tau_{\mathrm{tot}}} = \frac{p_{\mathrm{det}}}{\tau_{\mathrm{tot}}}. \tag{6.20}$$

Incident preparation, collection range, absorption and readout efficiencies, and no-response outcomes all enter p_{det}, while apparatus recovery and waiting enter the actual period. Detection efficiency and temporal response require calibration of the specific apparatus [29]. Light arriving at matter and matter leaving a record are successive stages of one process.

6.3 Records Carried by Light

Return to the illuminated stone. Different stone positions may produce different outgoing light from the same incident preparation. The distinction may enter propagation direction, spatial phase, arrival time, or polarization. Through collection, propagation, and imaging, a camera converts some of these distinctions into different pixel responses. The object's position can thus be read by apparatus far away.

Take two stone-position components A, B, first in a case where the outgoing light in each position branch is a pure state. After scattering, the components are correlated with $|\gamma_A\rangle$ and $|\gamma_B\rangle$, while their respective recoils and other outputs remain in the joint state. If the light states can be distinguished without error, light already carries the complete distinction between the two positions. If they overlap, the light state alone distinguishes them only partially.

Let $|m_A\rangle, |m_B\rangle$ denote the two retained orthogonal matter branches in this example, including recoil and other outputs that must still be tracked. The post-scattering joint state is

$$|\Omega\rangle = c_A|m_A\rangle|\gamma_A\rangle + c_B|m_B\rangle|\gamma_B\rangle, \tag{6.21}$$

Here $|c_A|^2 + |c_B|^2 = 1$. Position distinctions enter light through this joint correlation. If light–matter entanglement remains within each position branch, calculations must instead use the corresponding complete joint state. Conditional light states are then represented by density matrices; orthogonal supports allow light alone to carry an error-free position distinction.

While light travels from stone to camera, the stone continues moving. In the free-propagation example considered, outgoing light no longer recouples to retained

matter. Matter and the environment still participating in its action continue under the joint evolution U_{ret}, while light undergoes common free propagation U_γ. The two sides have states

$$\begin{aligned} |m_i(t)\rangle &= U_{\mathrm{ret}}(t)|m_i\rangle, \\ |\gamma_i(t)\rangle &= U_\gamma(t)|\gamma_i\rangle. \end{aligned} \tag{6.22}$$

The joint state before reception is therefore

$$|\Omega(t)\rangle = \sum_i c_i |m_i(t)\rangle|\gamma_i(t)\rangle. \tag{6.23}$$

Substituting into the inner product of the two light outputs and using $U_\gamma^\dagger U_\gamma = I$ gives

$$\begin{aligned} \langle\gamma_B(t)|\gamma_A(t)\rangle &= \langle\gamma_B|U_\gamma^\dagger U_\gamma|\gamma_A\rangle \\ &= \langle\gamma_B|\gamma_A\rangle. \end{aligned} \tag{6.24}$$

Propagation can convert phase differences into spatial differences on a receiving surface, making them accessible to suitable apparatus. Common unitary propagation does not increase the distinguishability of the states themselves. If light interacts with an intervening environment, the newly correlated environment must also be tracked. Collecting only some directions or ignoring degrees of freedom generally reduces the information available to the receiver.

The quantum instrument of Chapter 3 describes how a camera reads distinctions in light as pixels. Let F_r be the receiving effect acting on incident light, incorporating collection, imaging, and pixel response, with no response included as an outcome. Then

$$\sum_r F_r = I, \qquad p(r \mid i) = \langle\gamma_i(t)|F_r|\gamma_i(t)\rangle. \tag{6.25}$$

For ideal complete resolution, a pixel outcome r_A has probability one on branch A and zero on branch B, which gives another outcome. If the response satisfies the preservation criteria of Section 3.1, it forms a distant position record. When object positions are very close, the two outgoing light states may be difficult to distinguish within the actual collection range, preventing error-free resolution. Resolution depends on wavelength, collection range, propagation, and reception, rather than merely on the occurrence of light scattering.

The same receiving action determines the matter state after outcome r. Write

$$f_{ij}^{(r)} = \langle\gamma_j(t)|F_r|\gamma_i(t)\rangle, \qquad i,j \in \{A,B\}. \tag{6.26}$$

Apply the receiving action for r to $|\Omega(t)\rangle\langle\Omega(t)|$, then trace out the light and relevant auxiliary outputs after reception. Each $|\gamma_i\rangle\langle\gamma_j|$ gives $f_{ij}^{(r)}$, yielding the unnormalized matter state

$$\sigma_r = \sum_{i,j} c_i \overline{c_j}\, f_{ij}^{(r)} |m_i(t)\rangle\langle m_j(t)|. \tag{6.27}$$

Its trace is outcome probability p_r; when nonzero, σ_r/p_r is the state that continues propagating after this record. An ideal A record retains only branch A, while

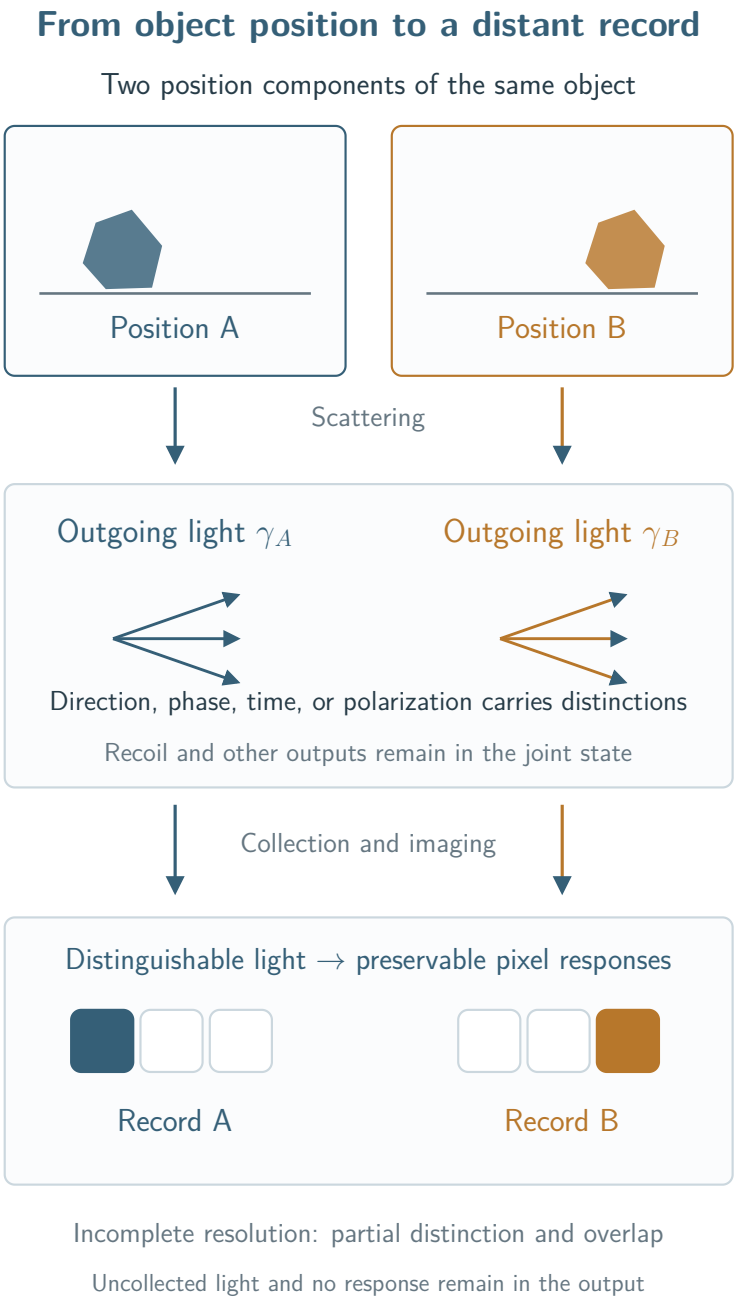


Figure 6-1. From object position to a distant record. At the top, two position components of the same object are correlated with different outgoing light. The middle retains correlations between light, recoil, and other outputs. At the bottom, receiving apparatus preserves the collected distinction in different pixel outcomes when the resolution conditions are met. Arrows connect physical processes without presupposing a point-by-point photon trajectory in transit. Uncollected light and no-response outcomes also belong to the complete output.

incomplete resolution retains the components and coherence permitted by the actual instrument.

Light can also leave the stone in directions not collected by the camera. Summing all camera outcomes, including no response, gives $\sum_r F_r = I$ and hence $\sum_r f_{ij}^{(r)} = \langle\gamma_j|\gamma_i\rangle$. Therefore

$$\sum_r \sigma_r = \sum_{i,j} c_i\overline{c_j}\langle\gamma_j|\gamma_i\rangle|m_i(t)\rangle\langle m_j(t)|. \tag{6.28}$$

This is exactly the partial trace over outgoing light of the joint state before reception. The cross term is multiplied by the light-output overlap: it vanishes for orthogonal light states and retains the corresponding coherence when they overlap. Reading only the stone, rather than the other matter outputs, requires tracing those out as well. Position distinctions already preserved in uncollected light remain even if no one receives it; switching off the camera afterward does not undo the earlier scattering. Camera records are the part of environmental records actually collected and preserved. Single-photon scattering experiments in atom interferometry have observed changes in interference contrast and recovered coherence in samples with restricted scattering correlations [30]. Such conditional recovery remains distinct from statistics summed over all outputs.

As the stone continues falling, later light scatters from its new state and may impart further recoil. Under the trajectory conditions of Section 5.5, successive images approximately show the same falling arc.

6.4 Light Signals and Spacetime Measurements

A camera receives light that left an earlier state of the object. The scattering time at the stone and the reception time at the camera are separated by propagation time; the reception time cannot simply be identified with the time when the distant object left the distinction carried by the light. Determining position and time of motion requires including light propagation in the same history.

A round-trip light signal gives a direct measurement process. Apparatus A is stationary in the uniform coordinate used, and its clock is calibrated to that time. It emits a narrow pulse at t_-; the pulse reaches a distant reflector B, returns to A, and leaves a reception record at t_+. These are effective emission and reception times corrected for apparatus delays, and reflection duration is negligible at the resolution used. If that duration is resolved, it is included in the round-trip history as an actual light–reflector interaction interval. With the long-wavelength, cycle-scale effective propagation of Section 6.1, take an approximately straight path with speed c_0 in both directions. If (R, t_B) is the reflection event, the two propagation legs satisfy, to this accuracy,

$$R = c_0(t_B - t_-), \qquad R = c_0(t_+ - t_B). \tag{6.29}$$

Equating the two expressions gives $2t_B = t_+ + t_-$; adding them gives $2R = c_0(t_+ - t_-)$. The distance and radar time of the reflection event are therefore

$$\boxed{\begin{aligned} R &= \frac{c_0}{2}(t_+ - t_-), \\ t_B^{\mathrm{rad}} &= \frac{t_+ + t_-}{2}. \end{aligned}} \tag{6.30}$$

Distance comes from the round-trip elapsed time, while the reflection time uses the symmetric round-trip radar assignment, connected to the definition of simultaneity used in light-signal synchronization [31]. If B carries its own clock, its reading at reflection must be compared with this assignment through actual synchronization and frequency calibration. Light connects actions at two locations; matter clocks participating in these actions supply the time readings.

Successive signals allow changes in target distance to be compared through successive reflection events. Each distance reading has an uncertainty range set by pulse width, propagation model, and apparatus time resolution. Velocity follows by comparing intervals between several events. Actual time-of-flight ranging reads distance from emission–reception time differences, while photon counting supplies individual arrival records [32].

Light also carries phase, enabling interference with a local receiver reference and comparison of frequency and phase changes. The matter timing reference uses the phase clock of Chapter 3. In the fixed-period, scalar-survival realization, let c_{ab} be the coherence of the normalized surviving state and $\omega_{ab} \neq 0$ the calibrated frequency difference. At cycle endpoints and in the corresponding effective continuous description,

$$c_{ab}(t) = c_{ab}(0)e^{-i\omega_{ab}t}. \tag{6.31}$$

At cycle N, unnormalized surviving coherence also contains the share factor q_T^N. This changes available statistics but, in this scalar return realization, does not change the normalized coherence phase. With nonzero coherence and continuously unwrapped phase, the phase change between two events is $\Delta\widetilde{\varphi}_{ab} = -\omega_{ab}\Delta t$, hence

$$\Delta t = -\frac{\Delta\widetilde{\varphi}_{ab}}{\omega_{ab}}. \tag{6.32}$$

Actual phase comparisons occur in the emitting and receiving apparatuses, and light carries phase relations between them. Both light's phase evolution in transit and the measurable propagation duration between apparatuses exist without assigning light a rest-frame clock of its own.

For the stone seen by the camera, the optical path and receiving delay likewise connect the image to the actual scattering event. In the uniform straight-path regime above, let L_γ be the calibrated path length from scattering to reception and t_{rec} the reception time corrected for apparatus delay. To the same effective accuracy,

$$t_{\mathrm{rec}} - t_{\mathrm{sc}} = \frac{L_\gamma}{c_0}, \qquad t_{\mathrm{sc}} = t_{\mathrm{rec}} - \frac{L_\gamma}{c_0}. \tag{6.33}$$

Reception time reports when light reaches the camera; subtracting propagation time identifies the earlier motion information it carries. Through a medium or nonuniform background, delays and curved paths must be recalibrated with the actual propagation model, adjusting distance and time readout accordingly.

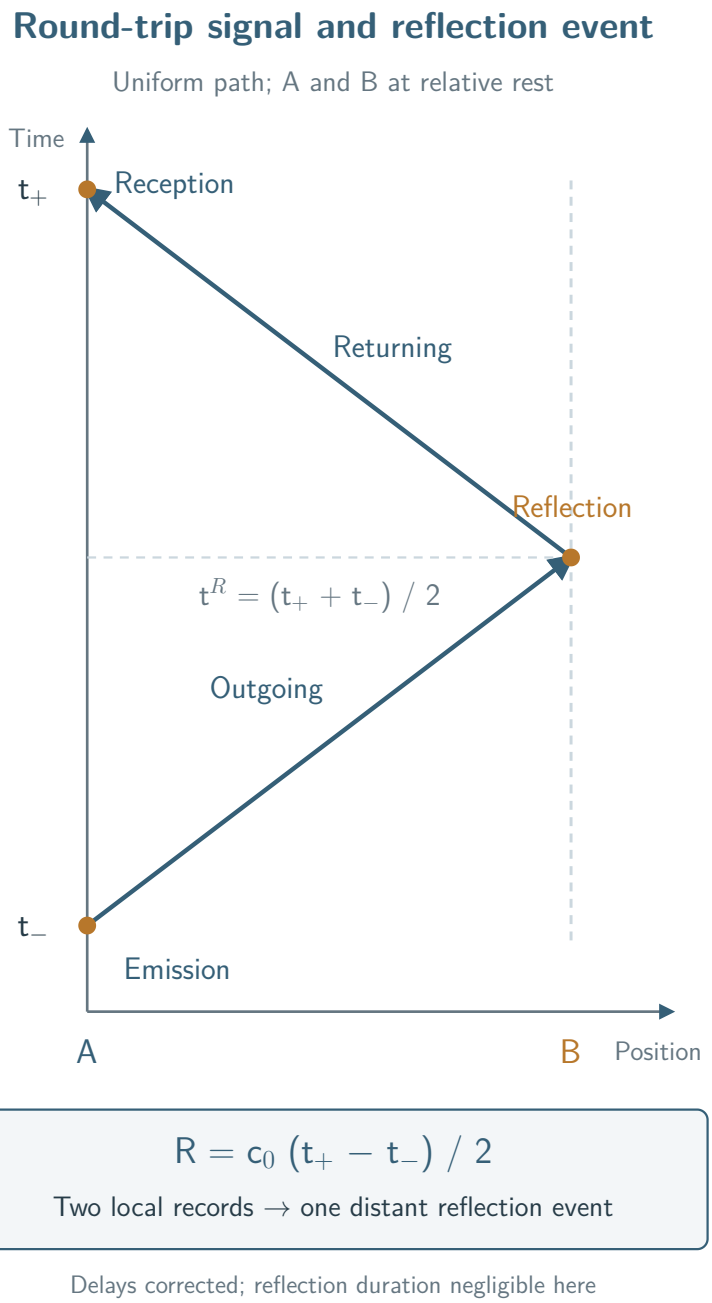


Figure 6-2. A round-trip signal and the reflection event. The vertical axis is calibrated time at A, and the horizontal axis is position in the same uniform coordinate. Here A, B are at relative rest. The outgoing and returning diagonal lines represent narrow-pulse centers propagating at c_0; the three points are emission, reflection, and reception events. Two local records at A give the reflection event's radar position and time. Apparatus delays have been corrected, and reflection duration is neglected at the illustrated resolution.

The stone continues moving after scattering, light continues propagating after leaving it, and the camera preserves a response after reception. These are successive physical processes in the same state space. Light allows distant motion to leave position and time records readable here; observed motion is established through these actual connections.

§7 Conclusions

Matter changes position while maintaining its structure and leaves records of motion through interactions with the environment and apparatus. Figure 7-1 connects propagation, exchange, and readout as one physical process.

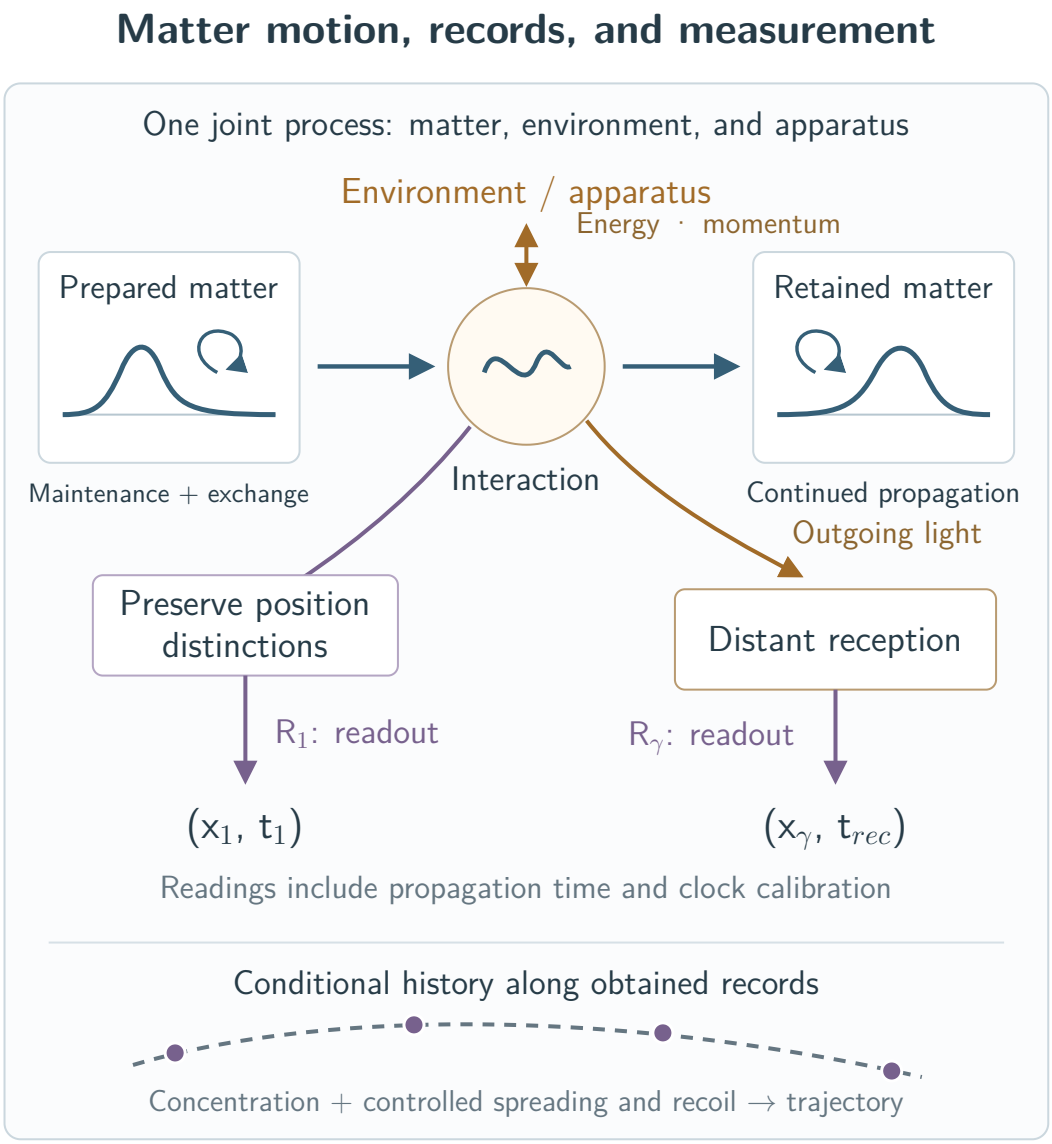


Figure 7-1. Matter motion, records, and measurement. Matter and all correlated outputs evolve jointly; records R are marked on readout arrows. Distant reception includes light-propagation delay. The lower dashed line indicates a conditional history continued along obtained records, approximating a trajectory when position is concentrated and spreading and recoil are controlled.

Matter delivered by formation and the accompanying degrees of freedom still involved in contact constitute a complete carrier. Internal maintenance and actual joint displacement act on this carrier. The intertwining relation

$$\widehat{H}\mathcal{J} = \mathcal{J}h_f \tag{7.1}$$

makes structural preservation compatible with evolving position amplitudes, while coherent exchange between neighbors composes global propagation. Under fixed-period, scalar-survival conditions and calibration by total elapsed duration, the long-wavelength, low-momentum expansion of a massive positive-energy branch gives

$$\boxed{M_{\mathrm{in},j}c_0^2 = E_{0j}.} \tag{7.2}$$

Spectral slope and curvature determine group velocity and dispersion. In the weak-binding example calculated, constituent motion and transverse current jointly make binding energy contribute to rest energy and collective inertia. The selected internal photon zero mode is determined by the kernel of the neutral response. Transverse propagation preserving this mode, under the common clock-and-length calibration, gives the long-wavelength spectrum $E_\gamma(k) \simeq c_0|k|$ and yields a pulse moving approximately at c_0.

Actual actions that change motion also carry exchanges. Apparatus, environment, exit outputs, and control work all enter the complete balance. For an energy-conserving joint action and a record resolving auxiliary input and output energies $\varepsilon_i, \varepsilon_o$, the corresponding conditional operator satisfies

$$[H_S, J_{oi}] = (\varepsilon_i - \varepsilon_o)J_{oi}, \tag{7.3}$$

which applies joint conservation to nonzero transitions. A single position label need not determine a unique energy difference.

The original shift, finite-difference momentum, and positivity of the state jointly constrain the position and momentum variances of the same prepared state. The continuum limit with controlled finite-wavelength and seam corrections recovers the uncertainty principle: the position and momentum distributions of the same prepared state cannot both be arbitrarily concentrated. This restriction holds before detection. The propagation components required for a localized preparation and their subsequent phase evolution determine, respectively, the uncertainty in the preparation and the spreading of the packet during propagation.

Complete propagation components also jointly determine other observable outputs. The two exponential components inside a barrier carry transmission current through matching at both ends, while static elastic action preserves incident energy. Double-slit phases and the overlap of all unread outputs jointly give the cross term

$$2\,\mathrm{Re}[A_1(x)A_2(x)^*\langle D_2|D_1\rangle]\,, \tag{7.4}$$

Complete path records make D_1, D_2 orthogonal, removing fringes from unconditioned screen statistics. A coherent equal-amplitude entangled pair gives $S = 2\sqrt{2}$ under the selected accessible readouts; complete local operations leave the other end's unconditioned statistics unchanged.

Outputs following records continue to propagate. The same charged-particle–atom interaction and three-dimensional propagation give forward correlations of two local excitations, providing a directional mechanism for successive tracks. A finite position instrument connects conditional narrowing, spreading between records, and recoil in one recursion. Narrow conditional histories maintained jointly by these actions allow the center of a low-speed object to approximately follow classical equations. The weak-background response of the bound-state example further shows

that rest energy after binding enters both inertia and gravitational potential and cancels from the leading-order acceleration.

A downward matter transition excites the light field through a closed loop, and coherent addition over creation times selects frequencies around the source energy difference.

Light carries position distinctions while exchanging energy and momentum; common free propagation preserves light-output overlaps. A reading at the receiver corresponds to an earlier scattering event; along the uniform straight optical path used here, reception and scattering are separated by L_γ/c_0. Spacetime comparisons additionally require calibrated apparatus responses and matter clocks. Records form through actual interactions; readout determines which preserved distinctions are obtained. Matter propagation, readable events, and continuous trajectories when their conditions hold are thereby connected within the same dynamics.

Appendix A. Supplementary Derivations of the Motion Mechanism

This appendix develops derivations used in Chapters 5 and 6: the contribution of the binding field to collective motion, successive position records, directional correlations between two atomic excitations, light-source excitation and spatial propagation, and error control for wave-packet uncertainty. Each section uses the physical identifications and common clock-and-length calibration of the main text, with $\hbar = 1$.

A.1 Binding Energy and Collective Motion

Consider the spherically symmetric, weakly electrostatically bound state of Chapter 5. Let c_m be the long-wavelength speed of the positive matter branch and $c_f = (\varepsilon_f \mu_f)^{-1/2}$ the speed of the same quadratic field action [12]. At fixed regularization, choose a simple, isolated, differentiable bound eigenvalue and use a low-velocity, conservative near-zone expansion, with self-interactions calibrated to the separated constituents. In terms of total momentum P and relative momentum p,

$$\begin{aligned} M &= m_1 + m_2, \quad \mu_r = \frac{m_1 m_2}{M}, \qquad \kappa = -\frac{q_1 q_2}{4\pi\varepsilon_f} > 0, \\ p_1 &= \frac{m_1}{M} P + p, \quad p_2 = \frac{m_2}{M} P - p, \\ H_{\mathrm{rel}} &= \frac{p^2}{2\mu_r} - \frac{\kappa}{r}, \quad I_b = \frac{\mu_r \kappa^2}{2}. \end{aligned} \tag{A.1}$$

Here I_b is the binding energy denoted by I in the main text. The ground state obeys $\langle p^2 \rangle = 2\mu_r I_b$ and $\langle \kappa/r \rangle = 2I_b$. Collective inertia is defined by the curvature of the actual eigenvalue $E_C(P)$. Write $H_i = \partial_{P_i} H$, $H_{ij} = \partial_{P_i}\partial_{P_j} H$, $Q = 1 - |\psi\rangle\langle\psi|$, and $R_\perp = Q(H - E_C)^{-1} Q$. Differentiating the eigenvalue equation gives

$$\begin{aligned} Q\partial_i \psi &= -R_\perp H_i \psi, \\ \partial_i \partial_j E_C &= \langle H_{ij} \rangle - \langle H_i R_\perp H_j + H_j R_\perp H_i \rangle. \end{aligned} \tag{A.2}$$

The zeroth-order action is $Mc_m^2 + P^2/(2M) + H_{\mathrm{rel}}$, for which $H_i^{(0)} = 0$ at $P = 0$. The leading binding correction to the curvature therefore comes from the explicit P^2 terms of the correction; state mixing enters at higher order for the fixed spectral gap.

This correction requires both the constituent positive branch and the binding field. For the original two-branch action, put $O = c_m \boldsymbol{\sigma} \cdot p$ and $\Delta = mc_m^2$. Eliminate the negative branch and normalize the upper component u_{up} as $\varphi = (1 + O^2/(4\Delta^2))^{1/2} u_{\mathrm{up}}$. This gives

$$H_+ - \Delta = V + \frac{O^2}{2\Delta} - \frac{O^4}{8\Delta^3} - \frac{[O,[O,V]]}{8\Delta^2} + \cdots . \tag{A.3}$$

Let $u(p)$ embed normalized two-component states into the complete four-component positive branch, and let $\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}$, where σ_i are the Pauli matrices. The charge and current matrix elements of this branch are

$$
\begin{aligned}
u(p')^\dagger u(p) &= 1 - \frac{|p'-p|^2}{8m^2c_m^2} + \frac{i\boldsymbol{\sigma}\cdot(p'\times p)}{4m^2c_m^2} + \cdots, \\
u(p')^\dagger c_m \boldsymbol{\alpha} u(p) &= \frac{p'+p}{2m} + \frac{i\boldsymbol{\sigma}\times(p'-p)}{2m} + \cdots .
\end{aligned}
\tag{A.4}
$$

Under a common momentum shift, the charge correction is at most linear in P, while the magnetization current is independent of P. The first-order expectation of the linear terms vanishes in the spherical state. Thus the P^2 terms at this order come from the free quartic momentum correction and the product of the two convection currents. The former gives

$$
\begin{aligned}
\delta H_{\rm kin} &= -\frac{p_1^4}{8m_1^3c_m^2} - \frac{p_2^4}{8m_2^3c_m^2}, \\
(\delta H_{\rm kin})_{P^2} &= -\frac{p^2P^2 + 2(p\cdot P)^2}{4\mu_r M^2 c_m^2}.
\end{aligned}
\tag{A.5}
$$

The transverse-current term follows from the same quadratic field action. In the conservative near zone, with no additional incoming radiation, first eliminate the transverse field:

$$
\begin{aligned}
L_T &= \int \left[\frac{\varepsilon_f}{2}\dot{A}_T^2 - \frac{|\nabla\times A_T|^2}{2\mu_f} + j_T\cdot A_T \right] d^3x, \\
A_T &= \mu_f(-\nabla^2)^{-1} j_T, \qquad L_T^{\rm on} = \frac{1}{2}\int j_T\cdot A_T\, d^3x.
\end{aligned}
\tag{A.6}
$$

The inverse Fourier transform of $(\delta_{ij} - k_ik_j/k^2)/k^2$ is

$$
K_{ij}^T(r) = \frac{\delta_{ij}}{4\pi r} + \partial_i\partial_j\Big(-\frac{r}{8\pi}\Big) = \frac{\delta_{ij}+n_in_j}{8\pi r}, \qquad n = \frac{\mathbf{r}}{r}. \tag{A.7}
$$

Expanding $j = j_1 + j_2$ cancels the factor $1/2$ through the two cross terms. Take the cross term for opposite charges and apply the first-order Legendre transform $\delta H = -\delta L(v_a = p_a/m_a)$ to obtain the corresponding Hamiltonian:

$$
\begin{aligned}
L_T^{12} &= -\frac{\kappa}{2c_f^2 r}[v_1\cdot v_2 + (v_1\cdot n)(v_2\cdot n)], \\
H_T^{12} &= \frac{\kappa}{2m_1m_2c_f^2 r}[p_1\cdot p_2 + (p_1\cdot n)(p_2\cdot n)]_{\rm sym}, \\
(H_T^{12})_{P^2} &= \frac{\kappa}{2M^2c_f^2 r}[P^2 + (P\cdot n)^2].
\end{aligned}
\tag{A.8}
$$

These are the canonical momenta of the effective action after eliminating the field; they already include its response. Current correlations remain operator products and cannot be removed by replacing each current with its zero mean. Using $\langle p_ip_j\rangle = \delta_{ij}\langle p^2\rangle/3$ and $\langle n_in_j/r\rangle = \delta_{ij}\langle 1/r\rangle/3$, the two contributions to collective kinetic energy are

$$
\begin{aligned}
\delta E_{\rm kin}|_{P^2} &= -\frac{5I_b}{6M^2c_m^2}P^2, \qquad \delta E_T|_{P^2} = \frac{4I_b}{3M^2c_f^2}P^2, \\
\frac{1}{M_{\rm in}} &= \frac{1}{M} + \frac{I_b}{M^2}\left(-\frac{5}{3c_m^2} + \frac{8}{3c_f^2}\right) + \mathcal{R}_B.
\end{aligned}
\tag{A.9}
$$

Here $\mathcal{R}_B$ collects higher-order binding, lattice, and field-elimination remainders; the effective window requires them to be smaller than the retained binding correction. With the actual common calibration $c_m = c_f = c_0$, expanding the reciprocal gives

$$\begin{aligned} E_C(P) &\simeq Mc_0^2 - I_b + \frac{P^2}{2M}\left(1 + \frac{I_b}{Mc_0^2}\right), \\ M_{\mathrm{in}} &\simeq M - \frac{I_b}{c_0^2}, \qquad M_{\mathrm{in}}c_0^2 \simeq E_C(0). \end{aligned} \tag{A.10}$$

Now use the weak-background deployment of Chapter 5: maintenance frequencies are multiplied by $1+\Phi$, while displacement coefficients and electrostatic strength are held fixed. The internal action becomes $H_{\mathrm{rel}}(\Phi) = T/(1+\Phi) + V_C$. Rescaling the bound-state coordinates by $1+\Phi$, or applying the Hellmann–Feynman relation to this action, gives

$$\begin{aligned} I_b(\Phi) &= (1+\Phi)I_b, \\ E_C(0,\Phi) &\simeq (1+\Phi)(Mc_0^2 - I_b), \\ \partial_\Phi E_C(0,\Phi)|_0 &\simeq E_C(0). \end{aligned} \tag{A.11}$$

The binding contribution thus enters the rest energy, motion curvature, and local passive background response of the same composite. The last relation uses the stated fixed connections; this calculation does not separately determine the composite body's active gravitational source action.

A.2 Position Records and Stable Conditional Width

Take a body position X, a movable reference position R, and a prepared pointer $|\eta\rangle$ on a finite position ring. Each round uses a freshly and independently prepared pointer; old records are retained without coupling back. Measure pointer addresses in lattice units and let S shift the pointer. The joint unitary controlled by relative position and its actual readout are

$$\begin{aligned} U_D &= \sum_{x,r} |x,r\rangle\langle x,r| \otimes S^{x-r}, \\ K_y &= \langle y|U_D|\eta\rangle = \eta\big(y - (X-R)\big), \\ \sum_y K_y^\dagger K_y &= 1, \qquad p_y = \mathrm{Tr}(K_y \rho K_y^\dagger), \qquad \rho_y = \frac{K_y \rho K_y^\dagger}{p_y}. \end{aligned} \tag{A.12}$$

In the continuous window, positions, readouts, and widths are converted to physical lengths using the same lattice spacing ℓ, and p is the physical wave number. To preserve the original positive motion branch, connect the readout to the actual branch structure. For the body and reference, write $h_a(p) = \Delta_a\sigma_z + \beta(p)\sigma_x$, where $\Delta_a = m_a c_0^2$ and $\beta(p) = c_0 \sin(p\ell)/\ell$; the Pauli matrices act in the motion-orientation basis chosen here. Define

$$
\begin{aligned}
\theta_a(p) &= \arctan\frac{\beta(p)}{\Delta_a}, \qquad W_a = e^{-i\sigma_y\theta_a/2},\\
W_a^\dagger h_a W_a &= E_a(p)\sigma_z, \qquad E_a(p) = \sqrt{\Delta_a^2+\beta(p)^2},\\
W &= W_m(P)\otimes W_{M_R}(P_R),\\
U_{\rm band} &= WU_DW^\dagger, \qquad K_y^{\rm band} = WK_yW^\dagger.
\end{aligned}
\tag{A.13}
$$

The three actual stages are unfolding by $W^\dagger$, detection by U_D, and restoration by W. They preserve the original positive branch at the output without positive-branch postselection. Each stage must replace or compensate the corresponding generator. The exact measured coordinate is $\widetilde{D} = W(X-R)W^\dagger$. In a smooth envelope without wraparound, $W_aX_aW_a^\dagger = X_a - \sigma_y\theta_a'(P_a)/2$, with $|\theta_a'| \le 1/(m_ac_0)$. When pointer width is much larger than the lattice spacing and these coordinate-correction lengths for both body and reference, the reading is calibrated as relative position.

In these calibrated coordinates write $D = X-R$, $\mu_D = mM_R/(m+M_R)$, and $\pi = \mu_D(P/m - P_R/M_R)$. A real Gaussian pointer in the low-velocity continuous window gives

$$
\begin{aligned}
K_y &= (2\pi s)^{-1/4}\exp\left[-\frac{(y-D)^2}{4s}\right],\\
\int dy\, K_y\rho K_y^\dagger &= \int dk\,|\widehat{\eta}(k)|^2 e^{-ikD}\rho e^{ikD},\\
P &\longmapsto P-k, \quad P_R \longmapsto P_R+k, \quad \operatorname{Var} k = \frac{1}{4s}.
\end{aligned}
\tag{A.14}
$$

Here $s>0$ is the pointer position variance. The same readout therefore produces both position concentration and recoil of the two bodies. It preserves joint translation exactly on the finite ring; the additive momentum relations use the long-wavelength regime without wraparound. Let E_i, E_f be the initial and final positive-branch energies averaged over all outcomes, and $\Delta_\Sigma = (m+M_R)c_0^2$. With degenerate pointer addresses, the energy account of the three controlled stages is

$$
\begin{aligned}
\mathcal{W}_1 &= \Delta_\Sigma - E_i, \qquad \mathcal{W}_D = 0, \qquad \mathcal{W}_2 = E_f - \Delta_\Sigma,\\
E_f &= \int dk\,|\widehat{\eta}(k)|^2\langle E_m(P-k) + E_R(P_R+k)\rangle,\\
\mathcal{W}_{\rm tot} &= E_f - E_i \simeq \frac{1}{8\mu_D s}, \qquad T_{\rm cycle} = \tau + 2t_{\rm rot} + t_D.
\end{aligned}
\tag{A.15}
$$

The final work expression uses the quadratic spectrum. Here τ is the free-propagation interval, and $t_{\rm rot}, t_D$ are the actual rotation and detection durations, all taken positive; any additional preparation, record-storage, or readout durations remain part of the apparatus's total cycle. Thus τ is not the duration of the entire process.

For a conditional Gaussian state, let $a = \operatorname{Var} D$, $b = \langle\{\delta D, \delta\pi\}\rangle/2$, and $d = \operatorname{Var}\pi$, and write $\gamma = \tau/\mu_D$. Free propagation followed by multiplication by the same K_y gives, by Gaussian integration,

$$
\begin{aligned}
a_- &= a + 2\gamma b + \gamma^2 d, \qquad b_- = b + \gamma d, \qquad d_- = d, \\
a' &= \frac{s a_-}{a_- + s}, \qquad b' = \frac{s b_-}{a_- + s}, \\
d' &= d_- - \frac{b_-^2}{a_- + s} + \frac{1}{4s}.
\end{aligned}
\tag{A.16}
$$

Concentration and back-action are parts of the same recursion; the final term is precisely the momentum variance generated by the pointer above. To find the stable width, take a pure Gaussian $\psi(D) \propto e^{-z(D-\bar{D})^2/2 + i\bar{\pi}D}$, with $\operatorname{Re} z > 0$. Free propagation and detection successively give

$$
\begin{aligned}
z' &= \frac{z}{1 + i\gamma z} + \frac{1}{2s}, \\
u &= \sqrt{\frac{1 + \sqrt{1 + 64 s^2/\gamma^2}}{2}}, \qquad v = \sqrt{u^2 - 1}, \qquad z_\pm = \frac{1 \pm u \mp iv}{4s}, \\
\frac{z_N - z_+}{z_N - z_-} &= (1 + i\gamma z_+)^{-2N} \frac{z_0 - z_+}{z_0 - z_-}, \\
\left|(1 + i\gamma z_+)^{-2}\right| &= \frac{u - 1}{u + 1} < 1.
\end{aligned}
\tag{A.17}
$$

This fractional transformation preserves the right half-plane, so every initial width in this pure Gaussian family approaches z_+. Using $a = (2 \operatorname{Re} z)^{-1}$, $b = -\operatorname{Im} z/(2 \operatorname{Re} z)$, and $d = |z|^2/(2 \operatorname{Re} z)$ gives the fixed widths after a record and before the next one:

$$
a_* = \frac{2s}{1 + u}, \qquad b_* = \frac{v}{2(1 + u)}, \qquad d_* = \frac{u}{4s}, \qquad a_{-*} = \frac{2s}{u - 1}. \tag{A.18}
$$

Since $b_* > 0$, the freely evolving position variance increases between records and reaches a_{-*} at the next record. A record also changes the conditional center. With innovation $\epsilon = y - (\bar{D} + \gamma\bar{\pi})$, whose variance at fixed history is $a_- + s$,

$$
\bar{D}' = \bar{D} + \gamma\bar{\pi} + \frac{a_-}{a_- + s}\epsilon, \qquad \bar{\pi}' = \bar{\pi} + \frac{b_-}{a_- + s}\epsilon. \tag{A.19}
$$

A stable conditional width is therefore compatible with continued dispersion of the centers of different histories. While the maximum width between records and accumulated recoil stay within the resolution regime of Chapter 5, the same body can continue along actual records as a localized trajectory relative to the reference. This result uses the controlled process and Gaussian window specified above.

A.3 Directional Correlations in Atomic Excitation

The atom–projectile action in Chapter 5 retains the positions of both nucleus and electron. Let the atomic reduced mass be $\mu_A = mM_N/(m + M_N)$ and $a = (\mu_A\kappa)^{-1}$. Its spherical ground state and three real $2p$ states are

$$
\psi_0(\mathbf{r}) = \frac{e^{-r/a}}{\sqrt{\pi a^3}}, \qquad \psi_j(\mathbf{r}) = \frac{r_j e^{-r/(2a)}}{4\sqrt{2\pi}\, a^{5/2}}, \qquad \omega_{21} = \frac{3\mu_A\kappa^2}{8}. \tag{A.20}
$$

Differentiate $\int e^{-\lambda r} e^{i\mathbf{k}\cdot\mathbf{r}} d^3r = 8\pi\lambda/(\lambda^2+k^2)^2$ with respect to k_j and put $\lambda = 3/(2a)$. The actual transition form factor is

$$\begin{aligned} F_j(\mathbf{k}) &= \langle 2p_j | e^{i\mathbf{k}\cdot\mathbf{r}} | 1s \rangle = i d_A k_j f(k), \\ d_A &= \frac{256a}{243\sqrt{2}}, \qquad f(k) = \left[1 + \left(\frac{2ak}{3}\right)^2\right]^{-3}. \end{aligned} \tag{A.21}$$

Use the Fourier convention of the main text, $\widetilde{W}(\mathbf{Q}) = \int W(\mathbf{u}) e^{i\mathbf{Q}\cdot\mathbf{u}} d^3u$. Shifting the nuclear and electronic terms separately in the original interaction gives

$$\begin{aligned} \widetilde{W}_{j0}(\mathbf{Q}) &= -i\, 4\pi g d_A \frac{Q_j}{Q^2} [\epsilon f(\epsilon Q) + b f(bQ)], \\ \mathcal{S}(Q) &= \sum_j |\widetilde{W}_{j0}(\mathbf{Q})|^2 \\ &= (4\pi g d_A)^2 \frac{[\epsilon f(\epsilon Q) + b f(bQ)]^2}{Q^2}. \end{aligned} \tag{A.22}$$

Here $\epsilon = m/(M_N + m)$, $b = M_N/(M_N + m)$, and $g = 2\kappa$. Both positive terms decrease with Q, so $\mathcal{S}(Q)$ is strictly decreasing for $Q > 0$; the excitation threshold excludes $Q = 0$. For atomic total mass $M_A = M_N + m$ and collision reduced mass $\mu_{\mathrm{col}} = M_\alpha M_A/(M_\alpha + M_A)$, energy balance with recoil of both participants gives

$$\begin{aligned} \mathbf{v}_{\mathrm{rel}} \cdot \mathbf{Q} &= \omega_{21} + \frac{Q^2}{2\mu_{\mathrm{col}}}, \qquad \mathbf{v}_{\mathrm{rel}} = \frac{\mathbf{p}}{M_\alpha} - \frac{\mathbf{K}_A}{M_A}, \\ k' &= \sqrt{k^2 - 2\mu_{\mathrm{col}}\omega_{21}}, \\ Q^2 &= (k - k')^2 + 4kk' \sin^2(\vartheta/2). \end{aligned} \tag{A.23}$$

The last two relations use the relative wave number $k = \mu_{\mathrm{col}}|\mathbf{v}_{\mathrm{rel}}|$ and scattering angle ϑ in the collision center-of-mass frame. The energy threshold fixes the minimum transfer $k - k'$; further deflection reduces the transition strength.

For two atoms, let $V_i(t)$ be the interaction-picture operator under the same original propagation, and let Π_{fg} select the two excited states. The leading double-excitation amplitude follows from the time-ordered expansion:

$$\begin{aligned} \Psi^{(2)}_{fg}(T) = &- e^{-iH_0T} \Pi_{fg} \int_0^T dt_2 \int_0^{t_2} dt_1 \\ &\times [V_2(t_2)V_1(t_1) + V_1(t_2)V_2(t_1)]\Psi_{00}. \end{aligned} \tag{A.24}$$

In the narrow-band, weak-collision, heavy and slowly moving atomic-center regime of the main text, locally expand the original three-dimensional propagation in the far field for the first ordering. The phases along the source-to-$\mathbf{a}_1$ and $\mathbf{a}_1$-to-$\mathbf{a}_2$ legs give momentum transfer $\mathbf{q} = k_0\widehat{\mathbf{a}}_1 - k_1\widehat{\mathbf{a}_2 - \mathbf{a}_1}$ in the first interaction region. Integrating there extracts $\widetilde{W}_{f0}(\mathbf{q})$, and summing the final $2p$ states gives $\mathcal{S}(q)$. The angular factor of the first transition is therefore stronger when the two propagation legs are nearly aligned, giving the forward correlation described in the main text. The complete double excitation still depends on both interactions. The reverse ordering is retained as well; without an order record, amplitudes are added before their norm is taken.

The local far-field regime must include the long-range tail of the transition charge, rather than being judged solely by the atomic radius a. This establishes the directional correlation of two excitations; ionization and condensation that amplify it into visible droplets remain distinct stages, as described in the main text.

A.4 Light-Source Excitation and Spatial Propagation

The local light source in Chapter 6 uses the original closed-edge action [12]. For source parameters $m_s, \kappa_s > 0$, the high-to-low source energy difference, outgoing-edge strength, and actual creation amplitude are

$$\begin{aligned} E_s &= \sqrt{m_s^2 + 2\kappa_s^2}, \qquad \Delta = E_s - m_s, \qquad \nu = 2\kappa_s, \\ j &= \nu/2, \qquad s_s = \sqrt{\frac{1 - m_s/E_s}{2}}, \qquad A = \frac{\kappa_s s_s}{\sqrt{2}} > 0. \end{aligned} \tag{A.25}$$

Let r be the source return slot, h the original signed three-dimensional propagation in the outgoing register, and $f = (\ell\mathcal{C}|0, z\rangle, 0)$ the normalized circulation source, with $\mathcal{C}$ the original centered-difference curl. Joint propagation retaining the source return, and the leading outgoing amplitude, are

$$\begin{aligned} K &= \begin{pmatrix} 0 & -ij\langle f| \\ ij|f\rangle & h \end{pmatrix}, \\ \chi_T^{(1)} &= -i\lambda A\, P_\gamma \int_0^T e^{-iK(T-t)} r\, e^{-i\Delta t}\, dt. \end{aligned} \tag{A.26}$$

Here P_γ projects onto the outgoing register and λ controls the actual source-link strength. Other source outputs remain in the complete action. For a frequency component ε of K, the time integral is

$$\int_0^T e^{-i\varepsilon(T-t)} e^{-i\Delta t} dt = T e^{-i(\varepsilon+\Delta)T/2} \operatorname{sinc}\left(\frac{(\varepsilon - \Delta)T}{2}\right). \tag{A.27}$$

The finite action time therefore favors output near the source energy difference, with a main-lobe width of order T^{-1} for this spectral-selection factor. That frequency must lie in the spectrum actually reached by the source. The original parameters directly satisfy $0 < \Delta < \nu/\sqrt{2} < \nu$. In the local infinite-lattice propagation limit, the two nonzero-frequency branches of h are $\pm\nu|\boldsymbol{\sigma}_q|$, where $\boldsymbol{\sigma}_q = (\sin q_x, \sin q_y, \sin q_z)$. For the original circulation source normal to z, the positive-frequency density is

$$\rho_f(\varepsilon) = \int_{[-\pi,\pi]^3} \frac{\sin^2 q_x + \sin^2 q_y}{2} \delta(\varepsilon - \nu|\boldsymbol{\sigma}_q|) \frac{d^3q}{(2\pi)^3} > 0, \quad 0 < \varepsilon < \nu. \tag{A.28}$$

The energy surface is regular in this interval, and the source weight is strictly positive on an open part of it. Keeping the full return coupling j does not remove this spectral interval. With $m_f(z) = \langle f, (z - h)^{-1} f\rangle$, block elimination gives

$$\begin{aligned} \langle r, (z - K)^{-1} r\rangle &= \frac{1}{z - j^2 m_f(z)}, \\ \rho_r(\varepsilon) &= \frac{j^2 \rho_f(\varepsilon)}{[\varepsilon - j^2 \operatorname{Re} m_f(\varepsilon + i0)]^2 + [\pi j^2 \rho_f(\varepsilon)]^2} > 0. \end{aligned} \tag{A.29}$$

Thus the source energy difference lies in an accessible continuous band. A finite deployment must simultaneously retain the weak-source expansion, sufficiently dense spectrum, and an interval before boundary feedback; a fixed small lattice does not automatically realize this limit. The frequency selection acts on K, including returns. It neither discards negative frequencies, bound components, or other source outputs nor by itself identifies a finite-time output as a purely positive-frequency far field. Chapter 6 therefore retains the complete creation, propagation, and reception process.

The spatial correspondence is likewise fixed by the actual action. The original three-dimensional curl propagation preserves eight subspaces jointly defined by component and position parity. On the infinite lattice, or a periodic lattice with an even number of at least six sites in each direction, let $\mathbf{b} \in \{0,1\}^3$. Within one such subspace, component addresses obey

$$\begin{aligned} E_i:&\quad x \bmod 2 = \mathbf{b} \oplus e_i, \\ B_i:&\quad x \bmod 2 = \mathbf{b} \oplus (\mathbf{1} \oplus e_i), \\ x_{E_i} &= Ln + \ell(\mathbf{b} + e_i), \qquad x_{B_i} = Ln + \ell(\mathbf{b} + \mathbf{1} - e_i), \qquad L = 2\ell. \end{aligned} \tag{A.30}$$

Here $\oplus$ denotes componentwise addition modulo two and $\mathbf{1} = (1,1,1)$. Each curl step changes both a component and one coordinate parity, thereby preserving $\mathbf{b}$. These addresses are the edge and face positions of the coarse lattice. If C is its oriented edge-to-face incidence matrix, the original centered difference obeys $\mathcal{C}_{E\to B} = C/L$ and $\mathcal{C}_{B\to E} = C^\dagger/L$.

The leading linear action of the original field return [12] is $\dot{N} = -s_f b_f C^\dagger S$, $\dot{S} = s_f a_f C N$, with $a_f, b_f > 0$. Normalize by the quadratic field energy, setting $X = \sqrt{a_f} N$ and $Y = \sqrt{b_f} S$. Then

$$i\partial_t \begin{pmatrix} X \\ Y \end{pmatrix} = h_{\text{field}} \begin{pmatrix} X \\ Y \end{pmatrix}, \qquad h_{\text{field}} = i s_f \sqrt{a_f b_f} \begin{pmatrix} 0 & -C^\dagger \\ C & 0 \end{pmatrix}. \tag{A.31}$$

Let $J_{\mathbf{b}}$ place X at the addresses above and map Y to $-B$ in the original propagation. Let $\bar{\nu}$ be the propagation strength in the same actual time and H_γ^{prop} the three-dimensional propagation generator measured in that time. Equality of the action on each edge gives

$$\begin{aligned} s_f \sqrt{a_f b_f} = \bar{\nu}/2 \quad &\Longrightarrow \quad H_\gamma^{\text{prop}} J_{\mathbf{b}} = J_{\mathbf{b}} h_{\text{field}}, \\ e^{-itH_\gamma^{\text{prop}}} J_{\mathbf{b}} &= J_{\mathbf{b}} e^{-ith_{\text{field}}}, \\ c_f = s_f L \sqrt{a_f b_f} &= \bar{\nu}\ell = c_\gamma. \end{aligned} \tag{A.32}$$

This linear deployment therefore preserves propagation at every wave number. The coarse-lattice wave number $Q_i = 2k_i\ell$ obeys $|e^{iQ_i} - 1| = 2|\sin(k_i\ell)|$, connecting the two spatial representations through the same staggered positions. The coefficient equality must be satisfied by the actual action and clock-and-length calibration. It is not merely a unit conversion, nor does it identify the complete finite-flux state space with the single-pattern space. The main text uses this connection to calculate the group velocities and propagation delays of long-wavelength pulses in Chapter 6.

Along the x axis, use the spatial phase e^{ikx} and order transverse components as (E_y, E_z, B_y, B_z). The original curl generator restricts to

$$h_\perp(k) = \bar{\nu}\sin(k\ell)\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{A.33}$$

The first two rows connect E_y to B_z and E_z to $-B_y$, respectively; the last two give the inverse exchanges. Thus $(E_y + B_z, E_z - B_y)^T/\sqrt{2}$ and $(E_y - B_z, E_z + B_y)^T/\sqrt{2}$ have coefficients $+\bar{\nu}\sin(k\ell)$ and $-\bar{\nu}\sin(k\ell)$, respectively, giving the two-orientation representation used in Section 6.1. This fixed change of basis retains all four transverse components, with polarization corresponding to the two independent components within each orientation.

A.5 Error Control for Wave-Packet Uncertainty

Chapter 5 derives the finite-ring variance relation from the original shift and finite-difference momentum. Its continuous form requires control of both the wavelength correction and the seam contribution, together with a comparison between finite-difference momentum and the momentum of a continuous envelope.

First fix a Fourier branch of the finite ring with $k\ell_x \in [-\pi, \pi)$, and define the spectral momentum by $P_{\mathrm{sp}}|k\rangle = k|k\rangle$. The original finite-difference momentum remains $\Pi_x(k) = \sin(k\ell_x)/\ell_x$. If Q_δ projects onto the wave-number window $|k\ell_x| \le \delta$, the Taylor remainder gives

$$\|(\Pi_x - P_{\mathrm{sp}})Q_\delta\| \le \frac{\delta^3}{6\ell_x}. \tag{A.34}$$

The variances can be compared directly. For the same spectral probabilities w_k, any real function F satisfies

$$\mathrm{Var}[F(k)] = \tfrac{1}{2}\sum_{k,k'} w_k w_{k'}[F(k) - F(k')]^2. \tag{A.35}$$

The absolute derivative of $F(k) = \sin(k\ell_x)/\ell_x$ is at most one. Hence $\sigma_{\Pi_x} \le \sigma_{P_{\mathrm{sp}}}$, and the product bound in the main text also constrains the spectral momentum width. When the spectrum lies in the stated window and $\delta < \pi/2$, the mean value theorem further gives

$$\cos\delta\, \sigma_{P_{\mathrm{sp}}} \le \sigma_{\Pi_x} \le \sigma_{P_{\mathrm{sp}}}. \tag{A.36}$$

This connects the two momentum widths in the long-wavelength regime; the finite-ring commutator continues to retain the seam term of the main text.

A continuous envelope can be obtained directly by sampling. Take a normalized smooth, compactly supported vector envelope $\psi \in C_c^\infty(\mathbb{R}; \mathbb{C}^r)$, where r accommodates the retained internal components. Move the position origin to the middle of the ring and set

$$\begin{aligned} x_n &= \ell_x[n - (L-1)/2], \\ v_n &= \frac{\sqrt{\ell_x}\,\psi(x_n)}{\sqrt{Z_{\ell_x}}}, \qquad Z_{\ell_x} = \ell_x \sum_n \|\psi(x_n)\|^2. \end{aligned} \tag{A.37}$$

Let $\ell_x \to 0$ and $L\ell_x \to \infty$. Riemann sums give $Z_{\ell_x} \to 1$, and the fixed support is eventually separated from the seam. The seam term then vanishes, while neighboring amplitudes satisfy

$$\begin{aligned} 1 - \langle C_\ell \rangle &= \tfrac{1}{2} \sum_n \|v_{n+1} - v_n\|^2 \\ &= \frac{\ell_x^2}{2} \int \|\psi'(x)\|^2 \, dx + o(\ell_x^2). \end{aligned} \tag{A.38}$$

The Taylor expansion of the central difference and the Riemann sums give convergence of the first and second position and momentum moments and of their cross moment. Writing $\mathcal{E}_{\ell_x}\psi = (v_n)_n$ for the normalized sampling, the finite-difference action satisfies

$$\left\| \Pi_x \mathcal{E}_{\ell_x} \psi - \frac{\sqrt{\ell_x}}{\sqrt{Z_{\ell_x}}} \big(-i\psi'(x_n)\big)_n \right\| \longrightarrow 0. \tag{A.39}$$

The finite variance relation in the main text therefore tends to the continuous uncertainty relation with covariance retained. On these smooth envelopes, the limiting operators satisfy $[x, -i\partial_x]\psi = i\psi$. A normalized Gaussian envelope $\psi(x) \propto \exp[-(x - x_0)^2/(4\sigma^2) + ip_0 x]$ has $\sigma_X = \sigma$ and $\sigma_P = 1/(2\sigma)$, attaining the bound $1/2$; it may first be smoothly truncated and the truncation radius subsequently increased. This is an equal-time limit of state families and their moments, without requiring convergence to be uniform over arbitrary initial states and arbitrary durations.

During propagation, the original uniform action preserves the long-wavelength condition. For a fixed matter mode j, use the generator of Chapter 2 with active time s:

$$h_j = \mu_j Y \otimes I_x + g Z_{\mathrm{mov}} \otimes D_x, \qquad [h_j, C_\ell] = 0. \tag{A.40}$$

Thus $\varepsilon_k(s) = \varepsilon_k(0)$, while the seam population requires a separate estimate. Let $\partial = \{0, L-1\}$, let Q_d^∂ project onto positions whose shortest ring-graph distance from this set is at least the integer $d \geq 1$, and let $\eta_d = \mathrm{Tr}[\rho_0(I - Q_d^\partial)]$ be the initial total population of the seam buffer. With the maintenance term as the free part, the hopping term in the interaction picture still connects only neighboring positions, and its norm is at most $|g|$. Fewer than d hops cannot reach the seam from Q_d^∂. The Dyson expansion therefore gives

$$\begin{aligned} \|P_\partial e^{-ish_j} Q_d^\partial\| &\leq a_d(q), \qquad q = |g|\,|s|, \\ a_d(q) &= \sum_{m=d}^{\infty} \frac{q^m}{m!}, \qquad P_\partial = |0\rangle\langle 0| + |L-1\rangle\langle L-1|. \end{aligned} \tag{A.41}$$

Purifying the initial state, decomposing it with Q_d^∂, and applying the triangle inequality to the two parts yields

$$\begin{aligned} p_\partial(s) &:= p_0(s) + p_{L-1}(s) \\ &\leq \min\left\{1, \left[\sqrt{\eta_d} + \sqrt{1 - \eta_d}\, a_d(q)\right]^2\right\}, \\ \varepsilon_\partial(s) &\leq \tfrac{L}{2} p_\partial(s). \end{aligned} \tag{A.42}$$

The bound in the main text can thus be controlled using the initial wave-number distribution, distance to the seam, and active duration. This is a sufficient estimate with a nonzero tail. One must check that $Lp_{\partial}(s)$ is small; the mean position not having reached a particular location does not by itself provide that control.

In the fixed-period, scalar-survival realization of Chapter 2, the normalized surviving state at cycle endpoints obeys $s = f_{\mathrm{act}}t$, allowing the window above to be expressed in total elapsed duration. Within a cycle, if logical propagation is indeed generated by the same action with switching, the actual accumulated active duration $s_{\mathrm{act}}(t)$ must be used, giving $q = |g|s_{\mathrm{act}}(t)$. States during a general return process must be calculated from its complete joint action; the average active fraction alone supplies no error bound for those states.

References

[1] S.-W. Hu. *Dynamics of the Bost–Connes State Space*. Zenodo, v2, 2026. DOI: 10.5281/zenodo.23005935.

[2] S.-W. Hu. *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System*. Zenodo, v9, 2026. DOI: 10.5281/zenodo.23837043.

[3] W. H. Zurek. Pointer basis of quantum apparatus: Into what mixture does the wave packet collapse? *Physical Review D* **24** (1981), 1516–1525. DOI: 10.1103/PhysRevD.24.1516.

[4] E. B. Davies, J. T. Lewis. An operational approach to quantum probability. *Communications in Mathematical Physics* **17** (1970), 239–260. DOI: 10.1007/BF01647093.

[5] M. Ozawa. Quantum measuring processes of continuous observables. *Journal of Mathematical Physics* **25** (1984), 79–87. DOI: 10.1063/1.526000.

[6] N. F. Ramsey. A Molecular Beam Resonance Method with Separated Oscillating Fields. *Physical Review* **78** (1950), 695–699. DOI: 10.1103/PhysRev.78.695.

[7] P. Strasberg, G. Schaller, T. Brandes, M. Esposito. Quantum and Information Thermodynamics: A Unifying Framework Based on Repeated Interactions. *Physical Review X* **7** (2017), 021003. DOI: 10.1103/PhysRevX.7.021003.

[8] A. Tonomura, J. Endo, T. Matsuda, T. Kawasaki, H. Ezawa. Demonstration of single-electron buildup of an interference pattern. *American Journal of Physics* **57** (1989), 117–120. DOI: 10.1119/1.16104.

[9] D. L. Donoho, P. B. Stark. Uncertainty Principles and Signal Recovery. *SIAM Journal on Applied Mathematics* **49** (1989), 906–931. DOI: 10.1137/0149053.

[10] H. P. Robertson. The Uncertainty Principle. *Physical Review* **34** (1929), 163–164. DOI: 10.1103/PhysRev.34.163.

[11] E. Schrödinger. Zum Heisenbergschen Unschärfeprinzip. *Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse* (1930), 296–303. English translation: arXiv:quant-ph/9903100.

[12] S.-W. Hu. *The Mechanism of Fields from the Bost–Connes System*. Zenodo, v5, 2026. DOI: 10.5281/zenodo.23520706.

[13] N. F. Mott. The wave mechanics of α-ray tracks. *Proceedings of the Royal Society of London. Series A* **126** (1929), 79–84. DOI: 10.1098/rspa.1929.0205.

[14] R. Bach, D. Pope, S.-H. Liou, H. Batelaan. Controlled double-slit electron diffraction. *New Journal of Physics* **15** (2013), 033018. DOI: 10.1088/1367-2630/15/3/033018.

[15] B.-G. Englert. Fringe Visibility and Which-Way Information: An Inequality. *Physical Review Letters* **77** (1996), 2154–2157. DOI: 10.1103/PhysRevLett.77.2154.

[16] M. O. Scully, K. Drühl. Quantum eraser: A proposed photon correlation experiment concerning observation and "delayed choice" in quantum mechanics. *Physical Review A* **25** (1982), 2208–2213. DOI: 10.1103/PhysRevA.25.2208.

[17] Y.-H. Kim, R. Yu, S. P. Kulik, Y. Shih, M. O. Scully. Delayed "Choice" Quantum Eraser. *Physical Review Letters* **84** (2000), 1–5. DOI: 10.1103/PhysRevLett.84.1.

[18] G. Binnig, H. Rohrer, Ch. Gerber, E. Weibel. Surface Studies by Scanning Tunneling Microscopy. *Physical Review Letters* **49** (1982), 57–61. DOI: 10.1103/PhysRevLett.49.57.

[19] J. Tersoff, D. R. Hamann. Theory of the scanning tunneling microscope. *Physical Review B* **31** (1985), 805–813. DOI: 10.1103/PhysRevB.31.805.

[20] J. S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika* **1** (1964), 195–200. DOI: 10.1103/PhysicsPhysiqueFizika.1.195.

[21] J. F. Clauser, M. A. Horne, A. Shimony, R. A. Holt. Proposed Experiment to Test Local Hidden-Variable Theories. *Physical Review Letters* **23** (1969), 880–884. DOI: 10.1103/PhysRevLett.23.880.

[22] B. Hensen, H. Bernien, A. E. Dréau, et al. Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature* **526** (2015), 682–686. DOI: 10.1038/nature15759.

[23] E. Joos, H. D. Zeh. The emergence of classical properties through interaction with the environment. *Zeitschrift für Physik B Condensed Matter* **59** (1985), 223–243. DOI: 10.1007/BF01725541.

[24] T. Bhattacharya, S. Habib, K. Jacobs. Continuous Quantum Measurement and the Emergence of Classical Chaos. *Physical Review Letters* **85** (2000), 4852–4855. DOI: 10.1103/PhysRevLett.85.4852.

[25] P. Ehrenfest. Bemerkung über die angenäherte Gültigkeit der klassischen Mechanik innerhalb der Quantenmechanik. *Zeitschrift für Physik* **45** (1927), 455–457. DOI: 10.1007/BF01329203.

[26] A. H. Compton. A Quantum Theory of the Scattering of X-rays by Light Elements. *Physical Review* **21** (1923), 483–502. DOI: 10.1103/PhysRev.21.483.

[27] E. F. Nichols, G. F. Hull. The Pressure Due to Radiation. (Second Paper.) *Physical Review (Series I)* **17** (1903), 26–50. DOI: 10.1103/PhysRevSeriesI.17.26.

[28] R. J. Glauber. The Quantum Theory of Optical Coherence. *Physical Review* **130** (1963), 2529–2539. DOI: 10.1103/PhysRev.130.2529.

[29] S. V. Polyakov. Single-Photon Detector Calibration. In A. Migdall, S. V. Polyakov, J. Fan, J. C. Bienfang (eds.), *Single-Photon Generation and Detection: Physics and Applications*. Experimental Methods in the Physical Sciences, Vol. 45. Academic Press, 2013, Chapter 8, pp. 257–281. DOI: 10.1016/B978-0-12-387695-9.00008-1.

[30] M. S. Chapman, T. D. Hammond, A. Lenef, J. Schmiedmayer, R. A. Rubenstein, E. Smith, D. E. Pritchard. Photon Scattering from Atoms in an Atom Interferometer: Coherence Lost and Regained. *Physical Review Letters* **75** (1995), 3783–3787. DOI: 10.1103/PhysRevLett.75.3783.

[31] A. Einstein. Zur Elektrodynamik bewegter Körper. *Annalen der Physik* **322** (1905), 891–921. DOI: 10.1002/andp.19053221004.

[32] R. E. Warburton, A. McCarthy, A. M. Wallace, S. Hernandez-Marin, R. H. Hadfield, S. W. Nam, G. S. Buller. Subcentimeter depth resolution using a single-photon counting time-of-flight laser ranging system at 1550 nm wavelength. *Optics Letters* **32** (2007), 2266–2268. DOI: 10.1364/OL.32.002266.